

Overlap between reconstructed b -jets and hadronic τ -jets
in $HH \rightarrow b\bar{b}\tau^+\tau^-$:

Summary of objectives 3.1–3.4 and guide to the code

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Abstract

This report summarises the studies carried out on the overlap between reconstructed b -jets and hadronic τ candidates in ATLAS ntuples for the search for Higgs-boson pair production in the $HH \rightarrow b\bar{b}\tau^+\tau^-$ final state. For each of the objectives 3.1–3.4 it states the goal, describes the workflow and the scripts of the repository [giumont/overlap_resolver](#) used to address it, and collects the main results, pointing to where the complete set of plots can be found. The aim is to allow future reproducibility and extensions of the work.

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Chapter 1

Introduction and common conventions

1.1 Physics context and goal of the project

The latest public ATLAS result on the search for the production of a pair of Higgs bosons, with one boson decaying to $b\bar{b}$ and the other to $\tau^+\tau^-$, is reported in Ref. [1]. In this final state, the same detector activity can be reconstructed *at the same time* as a b -jet and as a τ_{had} candidate, and can be accepted by both the b -tagging and the τ identification algorithms.

The project addresses this problem in four steps, mirrored by the numeric prefix of the scripts in the repository:

Objective 3.1

Quantify the overlap between reconstructed b -jets and τ_{had} candidates, and compare their kinematic characteristics and identification discriminants (signal sample).

Objective 3.2

Quantify the classification quality of the discriminants, first separately for b -jets and τ_{had} , then by combining them “manually” in case of overlap (signal sample).

Objective 3.3

Improve the definition of the discriminants in the overlap region with a neural network (signal sample).

Objective 3.4

Study the separation of signal and background events based on the count of the objects selected with the two discriminants (signal and $t\bar{t}$ samples).

Chapters 2–5 follow the same structure: *Objective*, *Approach* (workflow, scripts and functions) and *Results and conclusions* (figures and where to find the others in the repository).

The Logbook referred to in the text is available as pdf (`logbook.pdf`) or in markdown (suggested format, `logbook.md`) on the repository and it contains additional, less organised notes.

At some point in discussion and at the end of each chapter, a summary collects the open issues and the suggested extensions, marked in [blue](#).

1.2 Data samples

All studies use ntuples (`AnalysisMiniTree`, campaign `mc23a`, “bypass noOR” mode) stored on CERN EOS, which are also part of the private repository (`/root_datasets/` directory); Table 1.1 summarises them.

Sample	Process	Files	Local folder
Signal	$HH \rightarrow b\bar{b}\tau^+\tau^-$, gluon-gluon fusion (GGF)	14	root_datasets/HH_bbtt/
Background	fully hadronic $t\bar{t}$	9	root_datasets/bkg_tt/

Table 1.1: Input samples. EOS paths: `/eos/user/c/cschiavi/HHBTT/NTUPLES/VBF-TRIG/GGF-mc23a-bypass` (signal) and `/eos/user/c/cschiavi/HHBTT/NTUPLES/VBF-TRIG/TTBAR-mc23a-bypass-no0R_mode` (background). To point the code to different files, change the input paths in `params.py`.

The ntuples consist of a single `TTree` with jagged per-event collections, read with `uproot` and manipulated with `awkward`. The branches used throughout the project belong to the following families: reconstructed jets (`recojet_antikt4PFlow_*`, anti- k_t $R = 0.4$ PFlow), reconstructed hadronic taus (`tau_*`), truth objects (`truthjet_*`, `truthtau_*`, ...) and analysis-level variables (`bbtt_*`).

1.3 Common conventions

Analysis-level objects. Only jets and taus flagged by the branches `recojet_antikt4PFlow_isAnalysisJet` and `tau_isAnalysisTau__NOSYS` are used (set in `params.py` through `JET_IS_ANALYSIS_BRANCH` and `TAU_IS_ANALYSIS_BRANCH`). Optional additional selections are available through `JET_SELECTION_MODE` (`all` or `btag85`, i.e. the ready-made 85% b -tagging working point `..._ftag_select_GN2v01_FixedCutBEff_85`) and `TAU_SELECTION_MODE` (`all` or `score85`, i.e. the same efficiency threshold on the τ identification score).

Geometric overlap. A (reconstructed jet, reconstructed τ) pair is *overlapping* if

$$\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2} < \Delta R_{\text{overlap}}, \quad (1.1)$$

with $\Delta R_{\text{overlap}} = 0.4$ (this choice is motivated in Chapter 2). Two pairing strategies are implemented:

- *all pairs*: every combination below threshold is kept, so that one object can appear in several pairs (`obj_3_1_overlap_kinematics.py`);
- *no repetitions*: a greedy global matching per event, which repeatedly takes the pair with the smallest ΔR , removes both objects and stops when the smallest residual ΔR exceeds the threshold (`obj_3_1_overlap_kinematics_no_reps.py`); each jet and each τ is in at most one pair, independently of the storage order.

Truth categories. Each pair is classified according to the truth of its two members: the jet is a true b -jet if `recojet_antikt4PFlow_HadronConeExclTruthLabelID == 5`, and the τ is a true hadronic τ if the truth flag `tau_truth_IsHadronicTau` is set (the repository also supports a geometric matching to truth taus, selected with `TRUTH_MODE_TAU`, but this options has never been used in this analysis). The four resulting categories are listed in Table 1.2; with `AGGREGATE_CATEGORIES = True` the categories `TT` and `FF` are merged in a single class `other` (but this option has never been used in this analysis).

Short	Key in the code	b -jet	τ	Name used in the notes
TF	<code>a_jet_true_tau_fake</code>	true	fake	“trust jet”
FT	<code>b_jet_false_tau_true</code>	fake	true	“trust tau”
TT	<code>c_jet_true_tau_true</code>	true	true	both real
FF	<code>d_jet_false_tau_false</code>	fake	fake	both fake

Table 1.2: Truth categories of the (jet, τ) pairs. The first letter refers to the jet (T = true b -jet), the second to the τ (T = true hadronic τ).

Discriminants. The discriminants used are:

- for b -jets, the GN2v01 tagger, available as a discrete *quantile* (`recojet_antikt4PFlow_ftag_quantile_GN`) values $1, \dots, 6$ corresponding to nominal efficiencies 97, 90, 85, 80, 75, 70% for the cut “quantile $\geq b$ ”; -1 means “not available”;
- for τ candidates, the GNTau score (`tau_GNTauScoreSigTrans_v0prune`) and the RNN scores `tau_RNNJetScoreSigTrans` (jet rejection) and `tau_RNNEleScoreSigTrans_v1` (electron rejection).

The τ scores are *signal transformed*: the distribution for true hadronic taus is flat in $[0, 1]$, so that a threshold x corresponds to a signal efficiency $\epsilon_\tau = 1 - x$ (see Chapter 3).

1.4 Preliminary exploration of the ntuples

The script `obj_3_0_root_file_analysis.py` lists all the branches grouped by family (using `params.BRANCH_PREFIX_GROUPS`), prints summary statistics (mean, standard deviation, range, NaN fraction, objects per event) and saves a histogram for each branch listed in `params.EXPLORATION_KEY_BRANCHES`; it works on both signal and background files. Older diagnostics (truth versus reconstructed labels, τ versus τ -jet) are in `src/other_code/`, and they are discussed in the `logbook.pdf` file attached on the repo. The main findings on the signal sample (275 946 events) that are needed in the following are:

- **Pre-selection.** 100% of the reconstructed τ and 99.5% of the reconstructed jets already satisfy the analysis flags (`tau_isAnalysisTau` and `recojet_isAnalysisJet`), so the collections are pre-filtered; the masks are nevertheless always applied. There are on average 2.04 τ and 5.69 jets per event; the τ multiplicity is consistent with $H \rightarrow \tau\tau$, while the jet multiplicity looks overestimated for a $b\bar{b}$ final state ([to be investigated](#)).
- **Jet truth flavour.** Among the analysis jets, `HadronConeExclTruthLabelID` is 0 (light) for 35.3%, 5 (b) for 31.9%, 15 (τ) for 30.8% and 4 (c) for 2.0%. $\sim 30\%$ of jets are labelled as τ ([the relationship between this ratio and the counts on hadronic taus could be investigated further, see logbook.pdf](#)).
- **Truth hadronic τ .** Only 72.9% of the reconstructed τ are true hadronic taus (`tau_truth_IsHadronicTau`), so this flag must be applied. The branch `tau_truthType` gives the same fraction for the value 1, and `tau_truthOrigin` is not populated; neither is used in the following.
- **Higgs-parent masks.** True hadronic taus come from a Higgs boson in 98.8% of the cases, as expected, and 96.8% of the jets with truth label 5 have a Higgs parent, but only 19.4% of the jets with label 15 do. [The reason is still unclear](#) (see Section 5.3.6); the two Higgs bosons are indexed in order of appearance, not according to their decay mode.

[The meaning of `tau\_tauTruthJetLabel` \(plausibly the truth flavour of the jet closest to the reconstructed \$\tau\$ \) is also still to be confirmed.](#)

1.5 Repository layout

Table 1.3 gives a map of the repository [giumont/overlap_resolver](https://github.com/giumont/overlap_resolver); the suggested order for a first pass is `obj_3_0` $\rightarrow$ `obj_3_1` $\rightarrow$ `obj_3_2` $\rightarrow$ dataset builder $\rightarrow$ notebooks. Except for `obj_3_0_root_file_analysis.py`, which takes the input file and an optional cap on the number of entries as arguments, all the scripts read their inputs and settings from `params.py`.

Path	Content
<code>src/params.py</code>	single source of truth for paths, branch names, thresholds and plot settings
<code>src/obj_3_0_*.py</code>	ntuple exploration (Section 1.4)
<code>src/obj_3_1_*.py</code>	overlap counting, pair kinematics, MET-based variables, scores (Chapter 2)
<code>src/obj_3_2_*.py</code>	classification quality of the taggers and score correlations (Chapter 3)
<code>src/overlap_pairs_dataset_builder.py</code>	builds the (jet, τ) pair dataset for the NN (Chapter 4)
<code>src/other_code/</code>	diagnostics, truth labelling/plotting helpers (<code>truth_vs_reco_params.py</code>), original C++ exploration
<code>notebooks/</code>	pair preprocessing, dense NN and transformer training (Chapter 4)
<code>outputs_datasets_analysis/</code>	curated results of the dataset studies (signal and $t\bar{t}$)
<code>outputs_training/</code>	results of the DNN and transformer trainings, including the evaluation on $t\bar{t}$
<code>root_datasets/</code>	input ntuples
<code>docs/</code>	logbook and this report, in .md/.txt and .pdf format

Table 1.3: Repository layout.

Practical notes. Requirements are `uproot`, `awkward`, `numpy`, `scipy` and `matplotlib`, plus a deep-learning framework for the notebooks. Console output and some plot labels are in Italian (e.g. “Coppie”, “Soglie Tau ID”). `obj_3_2_bjet_quantile_analysis.py`. The external `flavour_tagging` project on disk (`params.FLAVOUR_TAGGING_DIR`) is needed, can be downloaded using command:

```
git clone https://github.com/giumont/ML4FlavourTagging
cd ML4FlavourTagging
pip install -e .
```

Chapter 2

Objective 3.1 — Overlap between b -jets and τ -jets

2.1 Objective

On the signal sample, understand the correlation between the identification of b -jets and of τ_{had} candidates by (i) quantifying the overlap between the two types of reconstructed objects, (ii) comparing their kinematic characteristics and (iii) comparing the identification discriminants computed by the offline algorithms, separately for the four truth categories of Table 1.2. A second goal is to identify which quantities carry discriminating power, to be used later as inputs of the neural network (Chapter 4).

2.2 Approach

2.2.1 Workflow

The study is implemented in the scripts `obj_3_1_*.py` of `src/`; all settings (branch names, thresholds, plot ranges) live in `params.py`. The workflow is:

1. **Loading and selection.** The ntuples are read with `uproot` into `awkward` arrays and only analysis-level jets and taus are kept (helpers `load_files`, `get_analysis_selection`, `delta_r`, `summarize` and `section` of the shared utility module imported in the other files). Files missing a required branch are skipped with a warning.
2. **Definition of the reconstructed objects.**
 - b -jets: jets with `recojet_antikt4PFlow_ftag_select_GN2v01_FixedCutBEff_85 = 1` (85% working point of GN2v01).
 - τ candidates: taus with `tau_GNTauScoreSigTrans_v0prune  $\geq$  TAU_SCORE_WP85_THRESHOLD`. The threshold is the score giving 85% signal efficiency on all true hadronic taus at analysis level, computed from `sklearn.metrics.roc_curve` (see Chapter 3) so that the two taggers are compared at the same nominal efficiency.

Events without any selected b -jet or without any selected τ (“orphan” objects) are discarded from the pair studies.

3. **Geometric overlap.** For each event all (b -jet, τ) combinations are built with `ak.cartesian` (and unpacked with `ak.unzip`) and ΔR is computed with `delta_r` (`obj_3_1_geometric_overlap.py`). Three counting definitions are compared: the minimum ΔR for each reconstructed b -jet, the minimum ΔR for each reconstructed τ , and ΔR for all pairs in the event. The threshold $\Delta R_{\text{overlap}}$ is chosen from their behaviour (Section 2.3.1).

4. **Truth labelling.** Jets and taus are labelled as true/fake with `label_jets_and_taus` (module `truth_vs_reco_params.py` in `src/other_code/`), using `HadronConeExclTruthLabelID == 5` for jets and `tau_truth_IsHadronicTau == 1` for taus (`TRUTH_MODE_TAU = label;` geometric matches to truth taus instead). Pairs are then split in the four categories of Table 1.2.
5. **Pair construction.** Two variants are implemented: `obj_3_1_overlap_kinematics.py` keeps all pairs with $\Delta R < \Delta R_{\text{overlap}}$ (an object can appear more than once), while `obj_3_1_overlap_kinematics_no_reps.py` builds one-to-one pairs with the greedy matching: (1) compute ΔR for all pairs in the event, (2) take the pair with the smallest ΔR , (3) keep it if $\Delta R < \Delta R_{\text{overlap}}$ and remove both objects, (4) repeat on the remaining objects. The functions are `greedy_match_no_reps_indices` (returns local jet/tau indices per event), `build_matched_pair_features`, `categorize_matched_pairs`, `merge_pair_kinematics` and `check_no_repetition`, which verifies that no object is repeated. For all the main analysis, the former one is used, as the number of counts has shown only negligible differences between the two methods (see next section).
6. **Feature histograms per truth category.** For every variable of Table 2.1 a normalised histogram (`density=True`, controlled by `NORMALIZE_PAIR_HISTOGRAMS`) is drawn with one curve per category; ranges, binning and labels are set by the `*_VARIABLES_NO_REPS` and `VARIABLE_PLOT_CONFIG_NO_REPS` dictionaries, and the output goes to `params.OUTPUT_DIR_NO_REPS`. The same script also produces scatter plots of ΔR versus jet p_T and versus τp_T .

2.2.2 Variables studied

Table 2.1 lists the variables compared across the four categories. Two of them are derived quantities, implemented in `obj_3_1_overlap_met_tau.py` with `compute_tau_met_proj` and `compute_tau_mt`. With $\Delta\phi_{\tau, E_T^{\text{miss}}} = \phi_\tau - \phi_{E_T^{\text{miss}}}$ wrapped in $[-\pi, \pi]$:

$$MET_{\parallel, \tau} = E_T^{\text{miss}} \cos \Delta\phi_{\tau, E_T^{\text{miss}}}, \quad m_T = \sqrt{2 p_T^\tau E_T^{\text{miss}} [1 - \cos \Delta\phi_{\tau, E_T^{\text{miss}}}]}. \quad (2.1)$$

In the collinear approximation the neutrino of a true hadronic τ decay is aligned with the visible products, so that $\Delta\phi_{\tau, E_T^{\text{miss}}} \approx 0$, hence $MET_{\parallel, \tau} > 0$ and small m_T ; for a fake τ no such relation is expected. The scripts `obj_3_1_overlap_scores.py` and `obj_3_1_overlap_metadata.py` handle the identification scores and the pair metadata, respectively.

Variable	Branch / definition	Object
p_T, η, ϕ	<code>recojet_antikt4PFlow_pt__NOSYS, _eta, _phi; tau_pt__NOSYS, tau_eta, tau_phi</code>	jet, τ
Jet mass	<code>recojet_antikt4PFlow_m__NOSYS</code>	jet
Soft muons	<code>recojet_antikt4PFlow_n_muons__NOSYS</code>	jet
Prongs, decay mode	<code>tau_nProng, tau_decayMode</code>	τ
Charge	<code>tau_charge</code>	τ
$MET_{\parallel, \tau}, m_T$	derived (see above)	τ
$\Delta\eta, \Delta\phi, \Delta R$	angular separation of the pair	pair
$p_T^{\text{jet}}/p_T^\tau$	ratio of transverse momenta	pair
Primary vertices	<code>nPrimaryVertices</code>	event
b -tag quantile	<code>recojet_antikt4PFlow_ftag_quantile_GN2v01_Cg</code>	continuous
τ scores	<code>tau_GNTauScoreSigTrans_v0prune, tau_RNNJetScoreSigTrans, tau_RNNEleScoreSigTrans_v1</code>	τ

Table 2.1: Variables compared across the four truth categories in objective 3.1.

2.3 Results and conclusions

2.3.1 Size of the overlap and choice of the threshold

Figure 2.1 compares the three ΔR definitions. At small ΔR the three distributions coincide: every reconstructed object has exactly one close partner. This does *not* prove automatically that the two objects are the same physical object (in theory, two distinct but close objects are still possible); it only shows that there are never more than two. Based on this, the overlap threshold is fixed to

$$\Delta R_{\text{overlap}} = 0.4. \quad (2.2)$$

This value is dataset dependent and is used here as a cut to study the kinematics of the overlap in this sample.

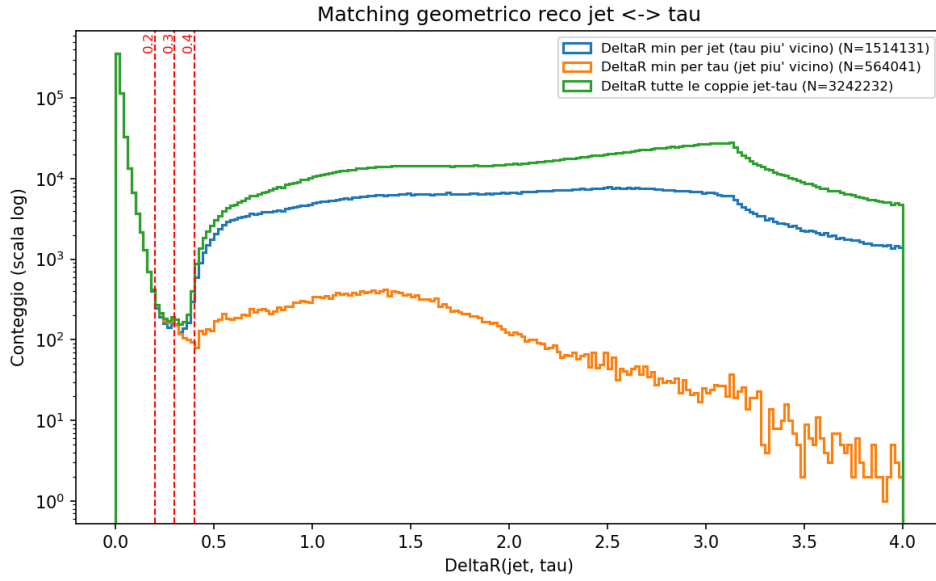


Figure 2.1: $\Delta R(\tau, b\text{-jet})$ for the selected objects: minimum over τ_{had} for each b -jet, minimum over b -jet for each τ_{had} , and all pairs in the event. see page 17/09 on `logbook.pdf` (“Scelta della soglia geometrica”); script `obj_3_1_geometric_overlap.py`.

Figure 2.2 shows ΔR for the four truth categories (normalised histograms, all reconstructed pairs). Correctly classified pairs (TT) are rare at small ΔR , as it is physically difficult to find a real b -jet and a real τ so close, while misclassified pairs concentrate in the overlap region: if the two taggers look at the same object, at least one of them must be wrong. Mis-tagged jets are more frequent than mis-tagged taus, which is expected because all combinations are counted and there are more taus than selected jets per event.

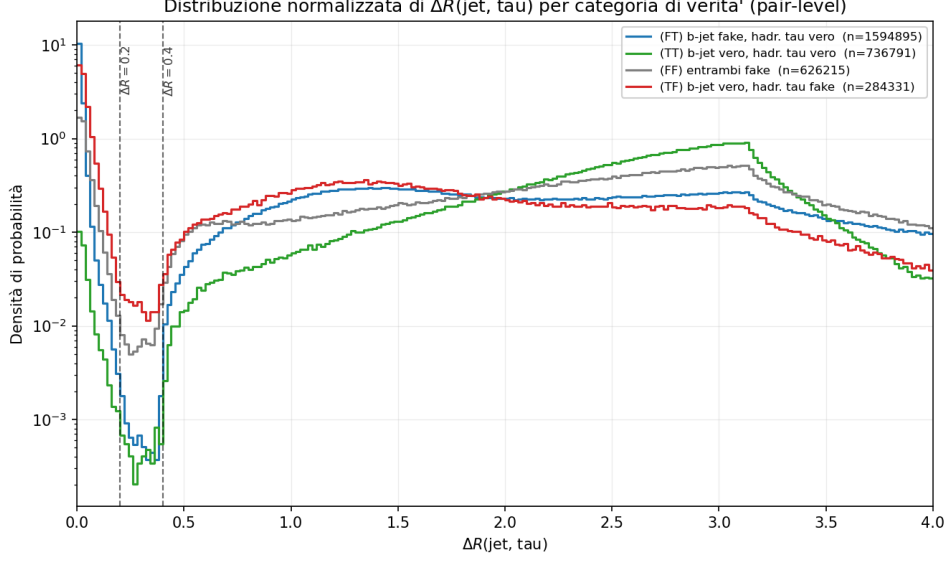


Figure 2.2: $\Delta R(\tau, b\text{-jet})$ for the four truth categories (all reconstructed pairs, normalised). See page 17/09 on `logbook.pdf` (“Gli overlapping di verità ...”).

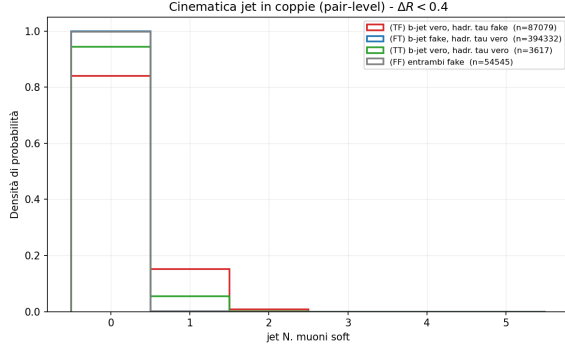
The robustness of the TT and FF populations was checked by comparing the two pairing strategies. The number of TT pairs in the signal sample is

$\Delta R_{\max}$	all pairs	no repetitions
0.4	3617	3587
0.2	3547	3546
0.045	2681	2681

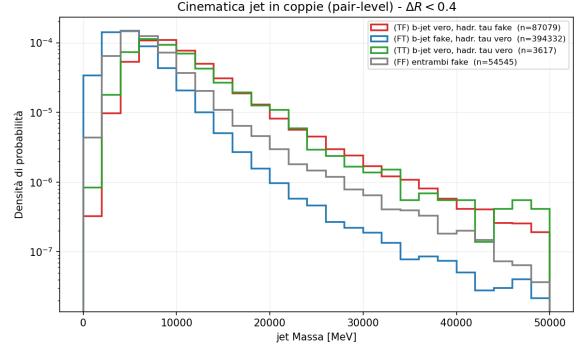
and similar results hold for FF: a sizeable fraction of these pairs is *irreducible*, i.e. it does not disappear when the pairing is made stricter. The TT class is a very small minority ($< 1\%$ of the signal taus considered in the score study, against $\sim 99\%$ for FT), which is the main statistical limitation of the project.

2.3.2 Kinematic characteristics of the overlapping objects

Jets. Only true b -jets (TT, TF) have a non-zero number of soft muons, and their mass peaks at larger values (5–10 GeV) than that of the fake jets (2–5 GeV), with longer tails; this is consistent with $m_b \gg m_\tau$. The FF category has an intermediate mass tail, compatible with a mixture of light jets, c -jets and gluon jets (Figure 2.3).



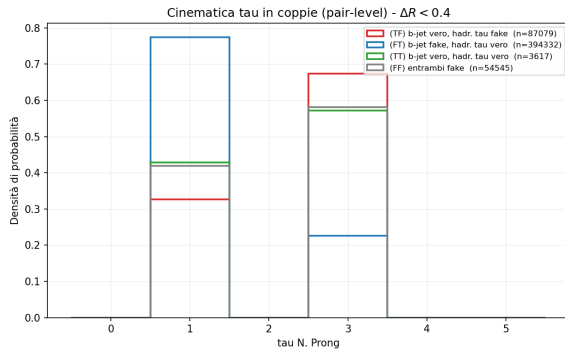
(a) Soft muons in the jet.



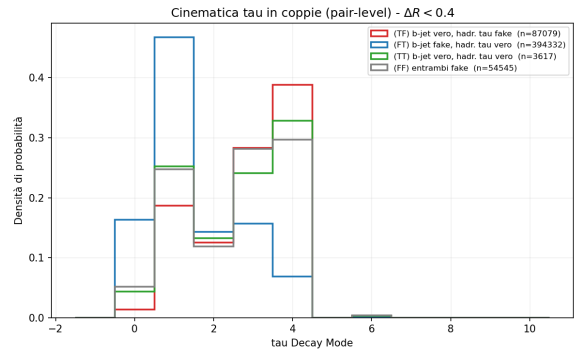
(b) Jet mass.

Figure 2.3: Jet variables for the four truth categories. See page 19/09 on [logbook.pdf](#) (“Potere discriminante di alcune caratteristiche dei jet”).

Taus. The number of prongs and the decay mode separate mainly FT from TF: b -jets mistaken for taus produce more prongs in the τ algorithm, since they are made of a multiplicity of tracks (Figure 2.4). The charge carries no discriminating power. [The interpretation of the decay-mode indices is still to be added.](#)



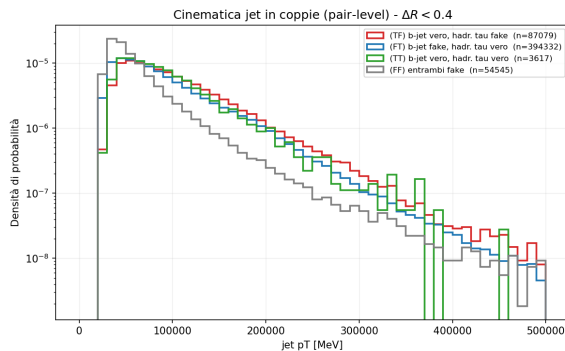
(a) `tau_nProng`.



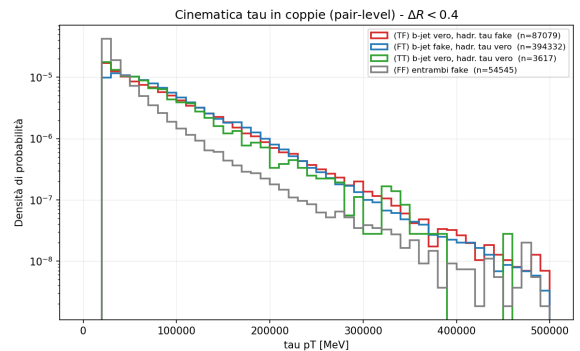
(b) `tau_decayMode`.

Figure 2.4: Reconstructed τ variables for the four truth categories. See page 19/09 on [logbook.pdf](#) (“Potere discriminante di alcune caratteristiche dei τ ”).

Transverse momentum. The p_T distributions of both objects are more peaked at low values (~ 10 GeV) for the FF category (Figure 2.5).



(a) Reconstructed jets.



(b) Reconstructed τ .

Figure 2.5: Transverse momentum of the objects in the four truth categories. Logbook 19/09.

MET-based variables. $MET_{\parallel,\tau}$ and m_T discriminate real from fake τ (T_ against F_) as expected from the collinear approximation: true taus concentrate at positive $MET_{\parallel,\tau}$ and small m_T (Figure 2.6).

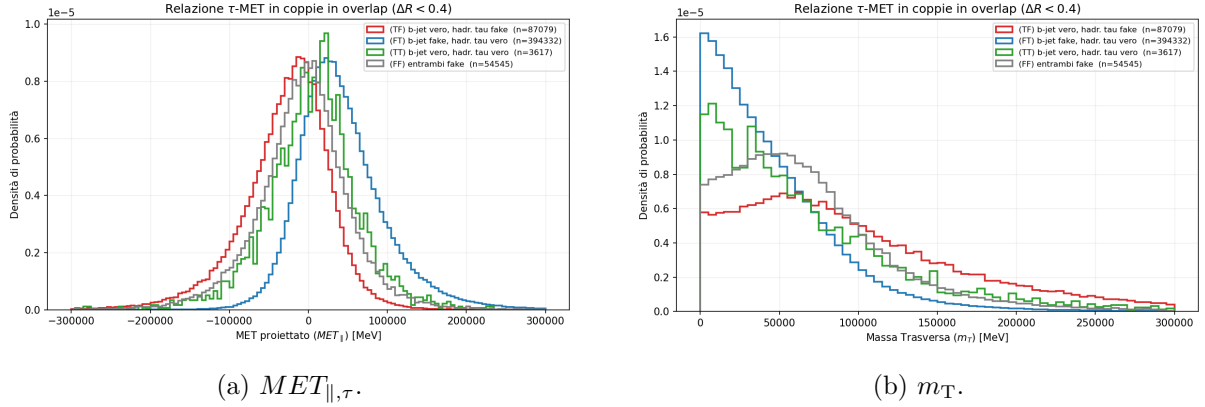


Figure 2.6: MET-based variables for the four truth categories. Logbook 20/09.

Angular variables and p_T ratio. The absolute pseudorapidities η_τ , η_{jet} are distinctive for the FF category, while ϕ carries no discriminating power. The relative variables $\Delta\eta$, $\Delta\phi$, ΔR and the ratio p_T^{jet}/p_T^τ are distinctive for FT (“trust tau”). A possible interpretation of $p_T^{jet} > p_T^\tau$ in the cases with a true jet is that the fake τ is reconstructed from a subset of the tracks of the real jet (Figure 2.7).

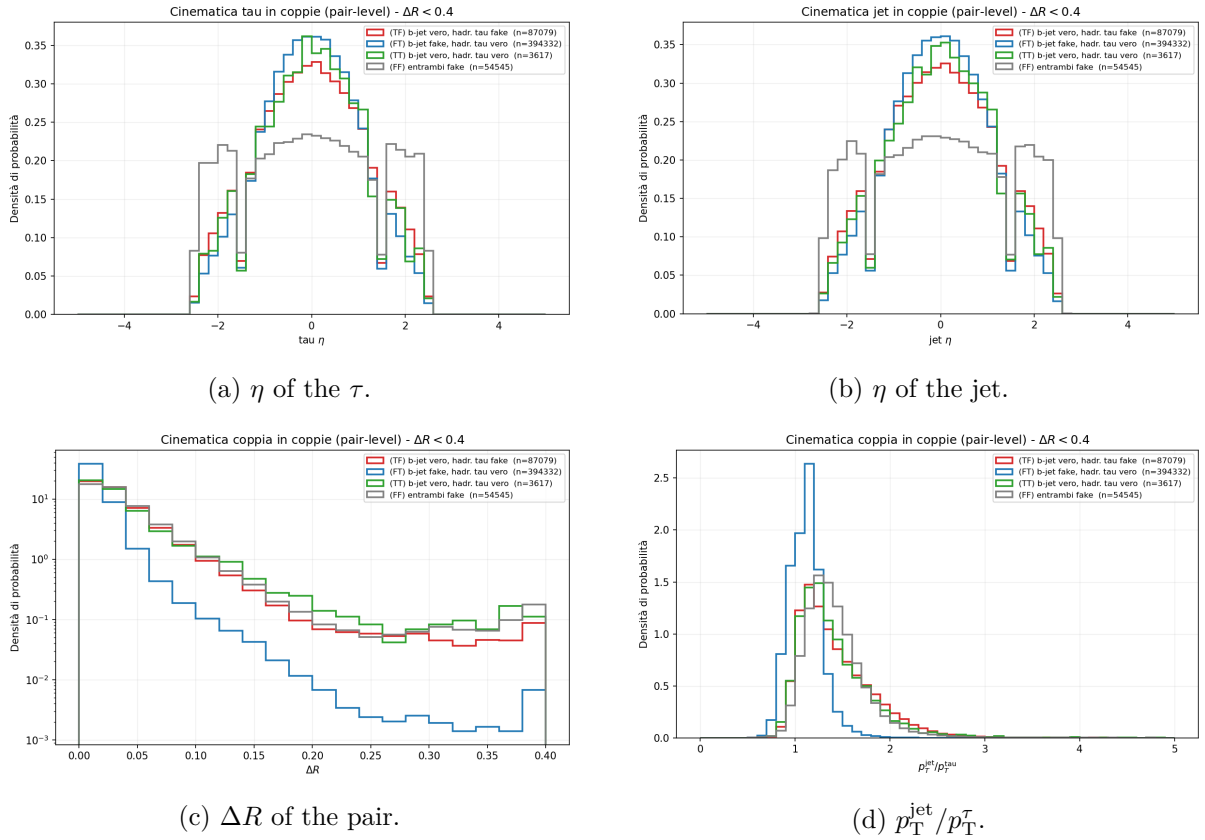


Figure 2.7: Angular variables and p_T ratio for the four truth categories. Logbook 20/09 and 21/09.

2.3.3 Identification scores in the overlap

The behaviour of the taggers for the four categories is discussed in detail in Chapter 3; the main outcomes relevant here are:

- the GN2 quantile separates jets with a true b -jet (T_-) from the others;
- the GNTau and RNN jet-rejection scores separate FT from TT, FF and TF, and are correlated (negatively, $\eta \simeq 0.4$, $\rho_s \simeq -0.4$) with the GN2 quantile essentially only in TT, which makes the joint information valuable;
- `tau_RNNEleScoreSigTrans_v1` behaves almost identically for TF, TT and FF and adds no complementary information.

2.3.4 Summary of the discriminating power

Table 2.2 summarises which variables separate which categories on the signal sample, and whether the separation is preserved on the $t\bar{t}$ background (Chapter 5). Variables *not* discriminating on the signal (ϕ_τ , ϕ_{jet} , `tau_charge`, `nPrimaryVertices`) are not used as inputs of the network.

Variable	Separates (signal)	Same behaviour on $t\bar{t}$?
Soft muons, jet mass	T_- vs F_-	yes
η_τ , η_{jet}	FF vs the rest	no
$MET_{\parallel,\tau}$, m_T	$_T$ vs $_F$	no
$p_T^{\text{jet}}/p_T^\tau$	FT vs the rest	no
<code>tau_decayMode</code> , <code>tau_nProng</code>	TF vs FT	inconclusive (low statistics)
p_T^τ , p_T^{jet}	FF vs the rest	inconclusive (low statistics)
$\Delta\eta$, $\Delta\phi$, ΔR	FT vs the rest	inconclusive (low statistics)
GN2 quantile	T_- vs F_-	inconclusive (low statistics)
τ scores (GNTau, RNN)	FT vs the rest	inconclusive (low statistics)

Table 2.2: Discriminating power of the variables on the signal sample (first letter = jet, second = τ ; “_” = any). The last column follows the notes on the $t\bar{t}$ background (see the plots in `outputs_datasets_analysis/outputs_tt/discriminance_analysis`).

2.3.5 Where to find the other plots

The figures above are a selection. The complete set of distributions (all variables, all categories) is in:

- `outputs_datasets_analysis/outputs_HH_bbtatau/discriminance_analysis/` (curated signal plots, including ϕ , `tau_charge`, `nPrimaryVertices`);
- `outputs_datasets_analysis/outputs_tt/discriminance_analysis/` (same variables on the $t\bar{t}$ background).

2.3.6 Conclusions and open points

- Reconstructed b -jet and τ candidates close in ΔR are mostly the result of a classification error: genuine TT pairs are rare.
- A number of kinematic and identification variables have a clear discriminating power on the signal and are used as inputs of the network; not all of them keep it on the background.
- *Open:* the exact composition of the FT overlap (how many fake jets have `HadronConeExclTruthLabelID == 15`, and how many have neither b nor τ flavour) and of the TT overlap (whether `tau_tauTruthJetLabel == 5` for all of them, which would support its interpretation as

truth flavour of the nearest jet) has not been studied yet; it matters because the network merges these sub-populations in a single class.

- *Open:* the meaning of `tau_tauTruthJetLabel` and the inconsistent propagation of `parentHiggsParentsMask` to jets with truth label 15 (see Section 1.4) are unresolved.

Chapter 3

Objective 3.2 — Quality of the classification

3.1 Objective

On the signal sample, quantify how well the identification discriminants perform, first *separately* for b -jets (GN2v01) and for τ_{had} candidates (GNTau, with the RNN taggers as comparison), and then study how the two discriminants behave *jointly* when a b -jet candidate and a τ candidate overlap, with the aim of combining them “manually”. The study answers three questions:

1. Do the nominal efficiencies of the GN2v01 quantile bins hold on this dataset?
2. What are the performance and the score distributions of the τ taggers, and how do they change with the truth category of the pair (Table 1.2)?
3. Is there a relation between the b -tagging quantile and the τ score in the overlap region, and does it differ between categories?

3.2 Approach

3.2.1 Workflow

The three scripts `obj_3_2_*.py` of `src/` share the configuration in `params.py` and the truth definitions of Table 1.2. The `flavour_tagging` project used for the plotting helpers is an external repository (path in `FLAVOUR_TAGGING_DIR`, `params.py`); it must be available on disk to run the b -jet script.

1. **b -jet quality: `obj_3_2_bjet_quantile_analysis.py`.** The branch `recojet_antikt4PFlow_ftag_quantile` is not a raw score but a discrete quantile assigned by ATLAS from a reference calibration: for bin $b \in \{1, \dots, 6\}$ the cut “quantile $\geq b$ ” has nominal b -jet efficiency 97, 90, 85, 80, 75, 70% (`JET_QUANTILE_BIN_NOMINAL_EFF`); the value -1 means “not available”. The script is organised in four parts:
 - *Part A*: properties of the quantile, overall and per subsample; real efficiency $P(\text{quantile} \geq b \mid \text{true } b\text{-jet})$ and fake acceptance per bin.
 - *Part B*: real versus nominal efficiency per bin, with the calibration plot `jet_gn2_quantile_calibration.png`.
 - *Part C*: independent check of the ready-made working point `recojet_antikt4PFlow_ftag_select_C`.
 - *Part D*: ROC curve, recalibrated working points for `TARGET_EFFICIENCIES` and the evaluation plots at the 85% working point (score distribution, background rejection versus efficiency, efficiency versus threshold, confusion matrix). The

helpers `plot_roc_curve` and `compute_working_points` come from the external `flavour_tagging` project (module `src.graphics`).

The truth label is `HadronConeExclTruthLabelID == 5`.

2. **τ quality:** `obj_3_2_tau_score_analysis.py`. The same metrics are computed for `tau_GNTauScoreSigTrans_v0prune`, taking as signal the true hadronic taus (`tau_isHadronicTau == 1`) and as background the fake ones. `sklearn.metrics.roc_curve` returns `tpr` (signal efficiency) and `fpr` (background efficiency) as a function of the score; the score thresholds for the target efficiencies [70, 75, 80, 85, 90, 97]% (the same family as the b -tagging working points) are the $(100 - \epsilon)$ percentiles of the true- τ score. The threshold at 85% is stored in `TAU_SCORE_WP85_THRESHOLD` and is the one used in Objective 3.1 to define the reconstructed τ candidates. The same analysis is repeated for the `tau_RNNJetScoreSigTrans` and `tau_RNNEleScoreSigTrans_v1` scores, and the score distributions are shown for the four truth categories of the overlapping pairs.
3. **Joint behaviour:** `obj_3_2_corr_scores_overlap.py`. For every (jet, τ) pair with $\Delta R < \text{DR_THRESHOLD_CORR}$ the b -tag quantile is related to the τ score. The pairs are split in the four truth categories and, for each, a violin plot shows the distribution of the τ score in every quantile bin, with the τ working points as dashed horizontal lines and a box reporting the number of pairs and the two indices below. Output files: `correlation_<score>_vs_btag_dr<DR>_truth_types.png` in `OUTPUT_DIR_CORR_SCORES` (`src/output/obj_3_2`).

[TODO: check that the function names listed above (and the parts A–D) match the current version of the scripts; the τ script is described here from the logbook, the README marks its scope as TODO]

3.2.2 Correlation indices

Since the b -tag quantile is a discrete variable, the standard Pearson coefficient is not appropriate. Two indices are used instead, for each truth category.

- The *correlation ratio* $\eta \in [0, 1]$, which treats the quantile X as categorical (so the bin -1 can be kept):

$$\eta^2 = \frac{\text{Var}_X(\mathbb{E}[s_\tau | X])}{\text{Var}(s_\tau)}, \quad (3.1)$$

where s_τ is the τ score. $\eta \simeq 0$ means that the mean score does not change across quantile bins (no dependence on GN2); $\eta \simeq 1$ means that the quantile almost fully separates the τ scores.

- The *Spearman coefficient* ρ_s , restricted to the valid bins ($X > 0$), which measures the monotonicity of the relation.

3.3 Results and conclusions

3.3.1 b -tagger: nominal versus real efficiency

The real efficiency of each GN2v01 quantile bin on the signal sample is compared with the nominal one in ???. The preliminary result of the logbook is that real and nominal efficiencies are in very good agreement, including for the ready-made 85% working point.

[TODO: AGGIUNGERE IMMAGINE]

3.3.2 τ -tagger: working points and signal transformation

Table 3.1 reports the working points of GNTau. The score is *signal transformed*: the distribution of the true taus is uniform on $[0, 1]$ (Figure 3.1), so that a threshold x selects approximately a fraction $1 - x$ of the signal, i.e. $\epsilon_\tau \simeq 1 - x$ (for instance, the 85% threshold is 0.163). This is an expected feature of the variable and it is important for what follows: for a uniform population the score carries no extra information, and any departure from uniformity in a sub-population signals a change in the tagger behaviour.

Target ϵ_τ	Score threshold	Measured ϵ_τ	Bkg. efficiency	Bkg. rejection
70%	0.3241	70.001%	1.243%	80.46
75%	0.2705	75.001%	1.631%	61.30
80%	0.2172	80.003%	2.175%	45.97
85%	0.1631	85.000%	2.980%	33.56
90%	0.1083	90.003%	4.346%	23.01
97%	0.0274	97.000%	10.809%	9.25

Table 3.1: Working points of `tau_GNTauScoreSigTrans_v0prune` on the signal sample (all analysis-level taus; signal = true hadronic taus). Logbook 17/09.

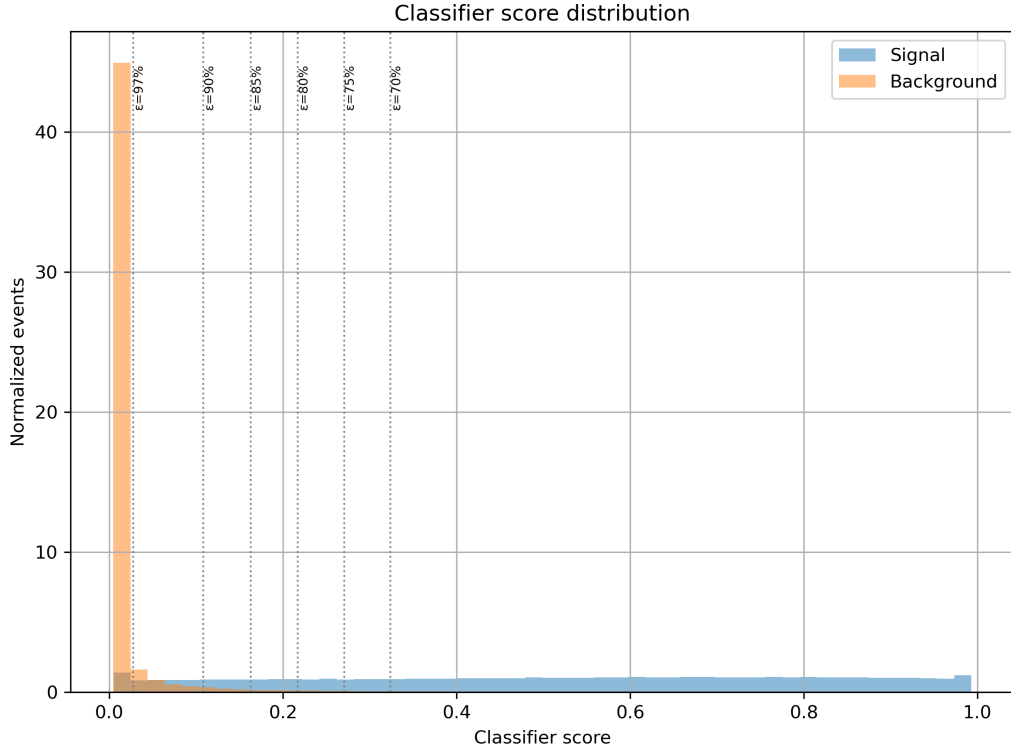
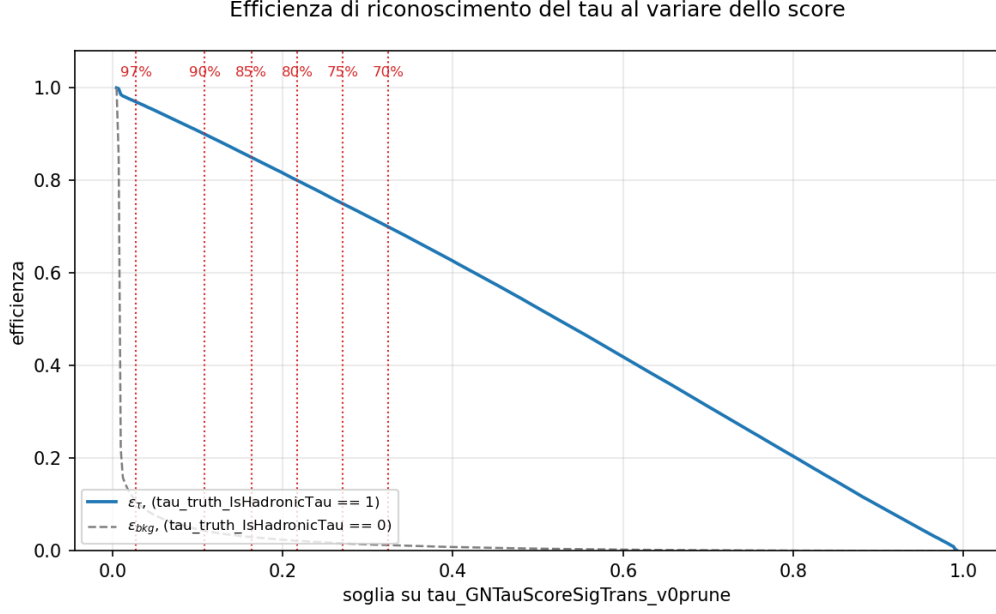
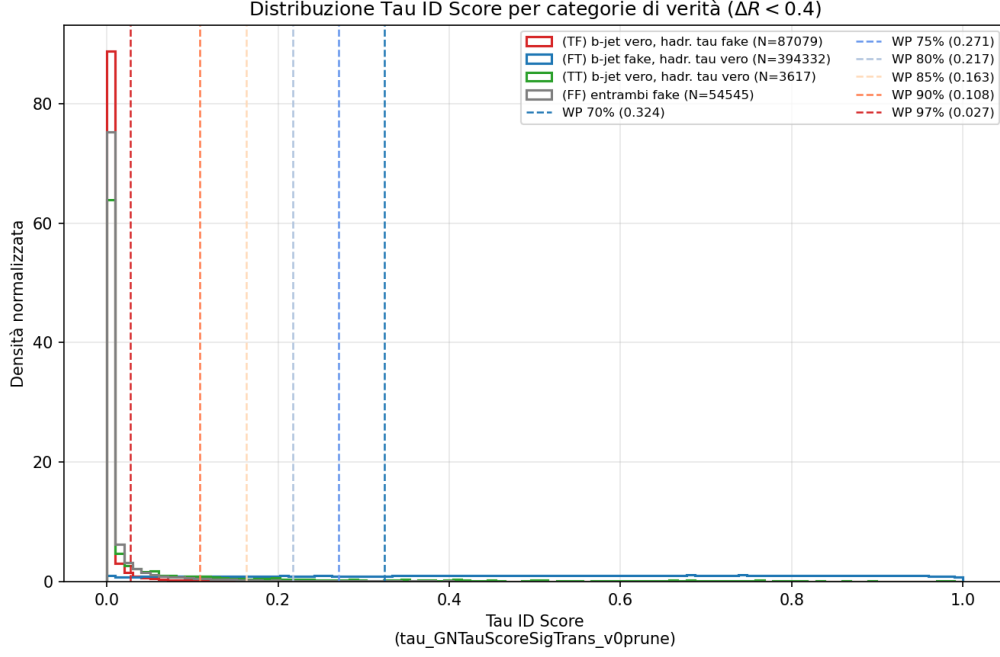


Figure 3.1: GNTau on the signal sample. (a) ROC curve with the working points of Table 3.1 (logbook 17/09). (b) The true- τ distribution is uniform (signal transformation; logbook 19/09). Script `obj_3_2_tau_score_analysis.py`; the remaining metrics (background rejection versus efficiency, efficiency versus threshold, confusion matrix at 85%) are in `outputs_datasets_analysis/outputs_HH_bbtatau/taggers_performance`.

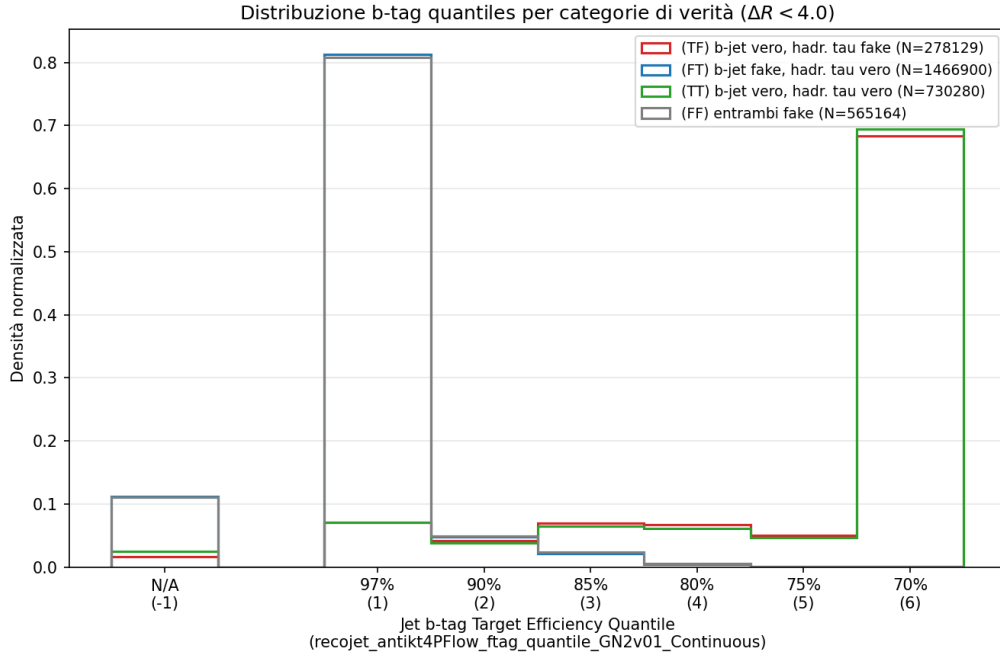
On the signal sample GNTau performs slightly better than the RNN tagger (e.g. AUC 0.9780 versus 0.9650 for `tau_RNNJetScoreSigTrans`) and the behaviour of the two as a function of the truth category is equivalent.

3.3.3 Tagger scores for the four truth categories

Figure 3.2 shows the scores of the two taggers for the overlapping pairs, split by category. The majority category **FT** (true τ ; $N = 394332$, about 99% of the true- τ pairs) follows the uniform distribution, as expected from the signal transformation. The very small **TT** category ($N = 3617$, $< 1\%$) has instead a peak at low scores (high efficiency region in the logbook wording), i.e. the tagger assigns a *lower* confidence to a real τ when it overlaps with a real b -jet. The same degradation is seen for GN2 in the **TT** case with respect to **TF**, but it is smaller. A plausible interpretation is that both networks are trained to reject background, and the additional energy deposits of the nearby object distort the features of the signal. The consequence is that the overlap has a larger impact on the confidence of the τ tagger than on that of the b -tagger. The scores of `tau_RNNJetScoreSigTrans` show the same behaviour.



(a) GNTau score.



(b) GN2 quantile.

Figure 3.2: Tagger outputs for the four truth categories of the overlapping pairs. Logbook 20/09 (“Performance dei tagger τ GNTau e b -jet GN2 ...”). The same plot for `tau_RNNJetScoreSigTrans` and `tau_RNNEleScoreSigTrans_v1` is in logbook 21/09.

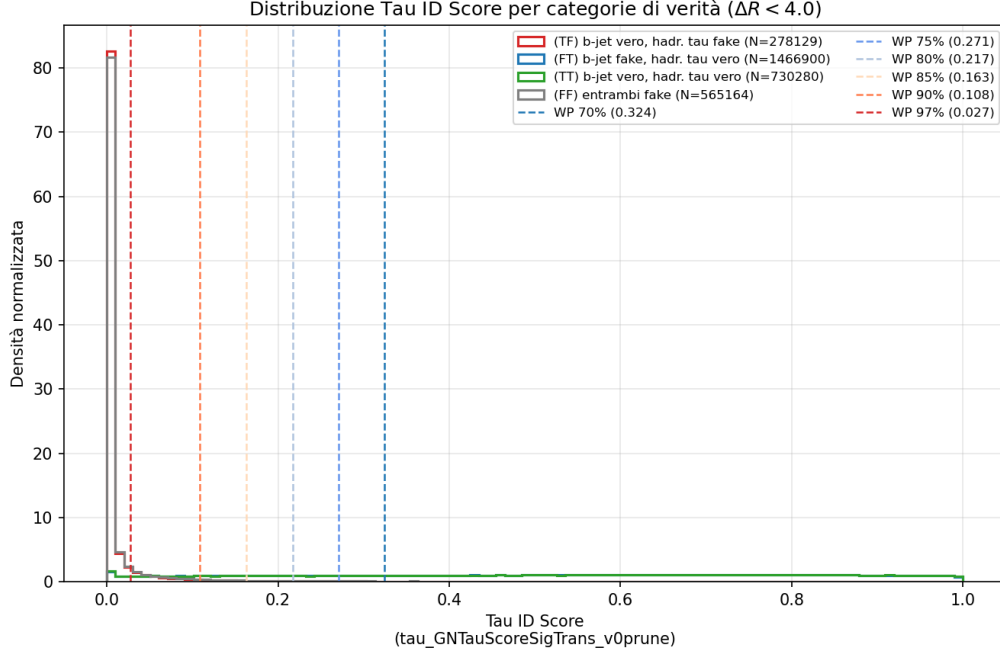
The electron-rejection score `tau_RNNEleScoreSigTrans_v1` behaves differently, because it is trained on a different problem (electrons versus hadronic taus). The true τ of FT is roughly uniform, but the fake τ of TF and FF are mostly not rejected, with a characteristic two-peak distribution (certain rejection or certain acceptance), and the same distribution is found for TT. It therefore gives no complementary information to separate the categories that the other two taggers do not already separate.

3.3.4 Joint information of the two discriminants

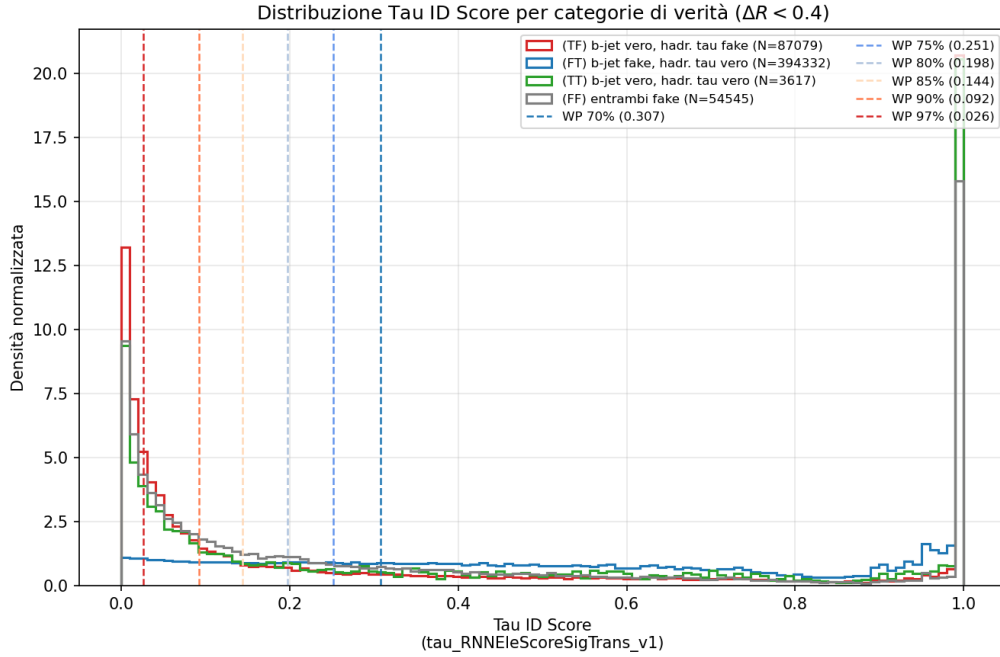
Table 3.2 gives the correlation indices between the τ score and the GN2 quantile for the pairs with $\Delta R < \text{DR_THRESHOLD_CORR}$, and Figure 3.3 the corresponding violin plots.

τ tagger	TT		TF		FT		FF	
	η	ρ_s	η	ρ_s	η	ρ_s	η	ρ_s
GNTau	0.413	-0.419	0.172	-0.112	0.066	+0.011	0.055	+0.012
RNNJet	0.386	-0.343	0.143	-0.020	0.087	-0.064	0.051	+0.020
RNNEle	0.134	-0.148	0.021	-0.053	0.073	-0.077	0.038	-0.025

Table 3.2: Correlation ratio η and Spearman coefficient ρ_s between the τ score and the GN2 quantile, per truth category. Logbook 21/09.



(a) GNTau versus GN2 quantile.



(b) RNNEle versus GN2 quantile (comparison).

Figure 3.3: Violin plots of the τ score in each GN2 quantile bin, for the four truth categories. Logbook 21/09; produced by `obj_3_2_corr_scores_overlap.py`. The RNNJet plot is in the same folder (`discriminance_analysis/results_with_reps/overlap_scores`).

The findings are:

- For FT (and FF) there is no correlation: the τ scores are uniform in every GN2 bin, as expected from the signal transformation.
- For GNTau and RNNJet the TT category shows a systematically stronger, negative correlation ($\eta \sim 0.4$, $\rho_s \sim -0.4$): the more a jet is clearly b -tagged, the lower the score of the

nearby τ candidate. The other categories have values close to zero.

- RNNEle is weakly correlated also for TT , consistent with its poor capability of separating the categories.

The joint behaviour of $[\tau \text{ score} \times \text{GN2 quantile}]$ therefore discriminates TT from FT , which motivates keeping at least one of GNTau and RNNJet together with GN2 as inputs of the network; RNNEle is the lowest priority of the three. In the end all the taggers were used as input features (Chapter 4).

3.3.5 Conclusions

- GN2v01 efficiencies agree with the nominal ones on this dataset.
- GNTau is slightly better than RNNJet; both are signal transformed and lose confidence on true τ that overlap with a true b -jet.
- The correlation of the two discriminants is significant only for TT , which makes it a useful feature for a combined classifier, but the TT statistics is very small.
- *Note:* no explicit rule for the manual combination of the two discriminants is derived here; it is delegated to the neural network of Chapter 4.

Chapter 4

Objective 3.3 — Neural network acting on the overlap

4.1 Objective

Develop a simple neural network that, for every overlapping (b -jet, τ) pair of the signal sample, predicts its truth category (TT, TF, FT, FF) from the kinematic variables and the tagger outputs studied in Objectives 3.1–3.2. The long-term goal is to improve the identification in the overlap region with respect to the two independent offline discriminants. The main difficulty is the strong class imbalance: TT is well below 1% of the pairs (Section 2.3.1), so the study concentrates on how to make the network sensitive to it.

4.2 Approach

4.2.1 Workflow

1. **Pair dataset:** `overlap_pairs_dataset_builder.py` builds one row per (reco jet, reco τ) pair within the overlap threshold, with its features and truth category (same categorisation as in Table 1.2). The pairing code and the variable extraction reuse the functions of Objective 3.1 (`obj_3_1_overlap_kinematics.py`, `obj_3_1_overlap_met_tau.py`).
2. **Preprocessing:** `notebooks/overlap_pairs_preprocessing.ipynb` produces the `.npz` files used for training. The preprocessing code was adapted from the earlier `flavour_tagging` repository (Objective 2.2). Two rules are essential:
 - the train/validation/test split is done *by event*: all the overlapping pairs of the same event are in the same subset, to avoid leakage between subsets;
 - mean and standard deviation for the feature standardisation are computed on the training set *only* and then applied to validation and test sets.
3. **Training and evaluation:** `notebooks/DNN_training.ipynb` (dense network); `notebooks/Transformer_training.ipynb` (alternative architecture, see the last section). Results are saved under `outputs_training/DNN/HH_bbtt_dataset/`, one folder per run, whose name encodes the configuration, e.g. `2layers_32_16_dropout0.1_LayerNorm_batch2048_lr0.001`.

4.2.2 Input features

The discriminance analysis (Table 2.2) was not actively used at the end to choose the input of the neural network, because also variables evaluated but found not to be discriminating (ϕ_τ , ϕ_{jet} , `tau_charge`, `nPrimaryVertices`) are used as features. The resulting input dataset contains 21 features per pair candidate. Seen the low input dimensionality no big issues are expected in

keeping them in the network, but [for coherence with previous analysis these additional variables could be removed from the training](#). Also, it was first considered to keep only GNTau as a τ tagger; in the end all the taggers were used as inputs.

Table 4.1: Input features and their ordering in the pair dataset, with the truth categories they discriminate on the $HH \rightarrow b\bar{b}\tau\tau$ sample (“T_” = true b -jet, “_T” = true τ ; “_” = not discriminating).

Index	Feature name	Discriminated classes (signal)
1	pair_dr	FT vs (TT, TF, FF)
2	pair_deta	FT vs (TT, TF, FF)
3	pair_dphi	FT vs (TT, TF, FF)
4	pair_pt_ratio	FT vs (TT, TF, FF)
5	jet_pt	FF vs (TT, TF, FT)
6	jet_eta	FF vs (TT, TF, FT)
7	jet_phi	–
8	jet_mass	(T_) vs (F_)
9	jet_n_muons	(T_) vs (F_)
10	jet_gn2_score	(T_) vs (F_)
11	tau_pt	FF vs (TT, TF, FT)
12	tau_eta	FF vs (TT, TF, FT)
13	tau_phi	–
14	tau_nProng	TF vs FT
15	tau_decayMode	TF vs FT
16	tau_charge	–
17	tau_gn_score	FT vs (TT, FF, TF)
18	tau_rnn_jet_score	FT vs (TT, FF, TF)
19	tau_rnn_ele_score	–
20	tau_met_proj	(_T) vs (_F)
21	tau_mt	(_T) vs (_F)

Several of these variables lose their discriminating power on the $t\bar{t}$ background (Chapter 5); this matters for the generalisation of the network beyond the signal sample.

4.2.3 Model and training setup

The baseline model is a dense network with two hidden layers ($32 \rightarrow 16$ units), dropout 0.1, layer normalisation, batch size 2048, learning rate 10^{-3} , weight decay 10^{-4} , 21 input features (Table 4.1) and four output classes. No resampling is applied (`sampling_strategy_none`): the imbalance is handled only through the loss.

The dataset contains 539 573 overlapping pairs, split by event into training (377 567), validation (80 923) and test (81 083) sets, i.e. approximately 70/15/15%. The class composition is the same in the three subsets within statistical fluctuations (Table 4.2); in particular TT is 0.65–0.76% of the pairs, with only 2 470 examples in the training set. After the standardisation, computed on the training set only, the means of the features in the validation and test sets are within ± 0.007 of zero and the standard deviations between 0.97 and 1.04. No NaN or infinite values are present.

4.2.4 Handling the class imbalance

The main reference for the discussion presented in this subsection is [2]; other useful resources are [4], [3], [3].

Cross-entropy for a sample with true class c and predicted probability p_c is $\text{CE} = -\log p_c$. It performs poorly under imbalance for two reasons: frequent classes dominate the loss, and it treats easy and hard examples alike. Two standard corrections were used and combined.

Table 4.2: Number of overlapping pairs per truth category in the training, validation and test sets (fraction of the subset in parentheses).

Class	Train	Validation	Test	Total
FF	38 153 (10.11%)	8 252 (10.20%)	8 140 (10.04%)	54 545
FT	276 062 (73.12%)	59 066 (72.99%)	59 204 (73.02%)	394 332
TF	60 882 (16.13%)	13 072 (16.15%)	13 125 (16.19%)	87 079
TT	2 470 (0.654%)	533 (0.659%)	614 (0.757%)	3 617
Total	377 567	80 923	81 083	539 573

- **Class weights** α_c inversely proportional to the class frequency in the training set,

$$\omega_c = \frac{N}{K N_c}, \quad (4.1)$$

with N the number of training pairs, $K = 4$ the number of classes and N_c the number in class c (for TT: $\omega_{TT} = 38.2$). This gives the balanced cross-entropy $-\alpha_c \log p_c$.

- **Focal loss**, with a modulating factor that reduces the weight of well classified examples, $FL = -\alpha_c(1 - p_c)^\gamma \log p_c$ (`CategoricalFocalLoss`); $\gamma = 0$ gives back the weighted cross-entropy.

In the scan the weights are $\alpha_c^{p_\alpha}$, with $0 < p_\alpha \leq 1$ (`CLASS_WEIGHTS = alpha**alpha_p`): $p_\alpha = 1$ is the full inverse-frequency weighting, smaller values flatten it.

4.2.5 Metrics and model selection

Precision = $TP/(TP + FP)$ and recall = $TP/(TP + FN)$ are computed per class; F_1 is their harmonic mean (see [6]). The *average precision* (AP) is the area under the precision–recall curve; `ap_macro` is its mean over classes and `val_ap_TT` its value for TT. The best epoch and the early stopping are based on `ap_macro` on the validation set, instead of the mean or median of the validation loss, because the loss is a poor absolute figure of merit here (see [2]).

4.3 Results and conclusions

4.3.1 Cross-entropy versus focal loss

Two trainings with the same hyper-parameters were compared: weighted cross-entropy and weighted focal loss with $\gamma = 2$ (the default value). Table 4.3 reports the test-set metrics and Figure 4.1 the corresponding confusion matrices.

The main findings are:

- With weighted cross-entropy the network *never* assigns a pair to TT under the `argmax` decision, even with $\omega_{TT} = 38.2$: precision and recall of TT are zero on the test set. Weighting alone is not enough to make TT win the comparison between the class scores.
- With the focal loss the network does predict TT: the test-set recall is 42.7% (and 45.9% on the training set, so it is not a training-set artefact).
- The gain has a cost. The TF recall drops by about 25 points (90.0% $\rightarrow$ 64.4%), the FF precision decreases, the accuracy goes from 94.8% to 87.7% and the macro- F_1 from 0.677 to 0.646. The focal loss therefore does not improve the classifier globally: it changes the trade-off between classes.
- The TT purity is very low (4.7%). From recall and precision, about 262 of the 614 true TT are found among $\sim 5.6 \times 10^3$ predicted TT, i.e. most of the TT predictions are false positives.

Class	Recall		Precision		F_1	
	CE	Focal	CE	Focal	CE	Focal
FF	82.76%	91.31%	81.28%	66.60%	0.8201	0.7703
FT	98.52%	92.80%	97.34%	99.01%	0.9793	0.9580
TF	89.96%	64.43%	91.75%	95.61%	0.9084	0.7698
TT	0.00%	42.67%	0.00%	4.69%	0.0000	0.0845

Table 4.3: Test-set metrics for the weighted cross-entropy (CE) and the weighted focal loss ($\gamma = 2$) runs. Logbook 23/09.

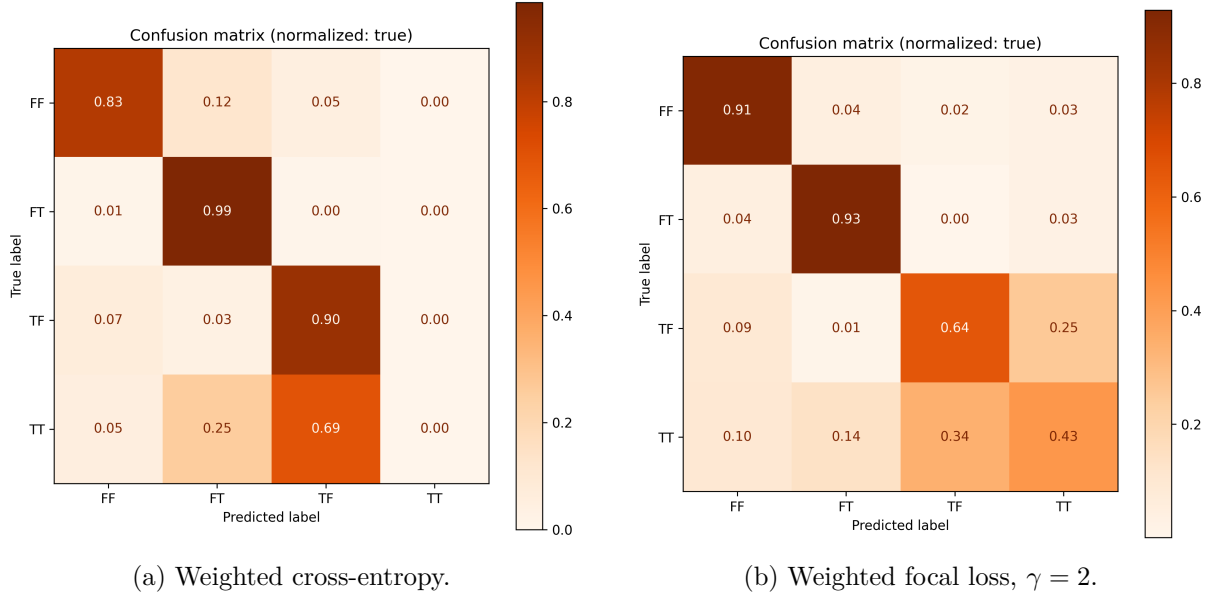


Figure 4.1: Test-set confusion matrices of the two runs of Table 4.3. Logbook 23/09; run folders in `outputs_training/DNN/HH_bbtt_dataset/` with `criterion_CrossEntropyLoss()` and `criterion_CategoricalFocalLoss()`. The training curves and the ROC/precision–recall plots of each run are in the same folders.

4.3.2 Scan of the focal-loss parameters

A grid scan over (p_α, γ) was run, selecting the best epoch on `ap_macro`. The two validation metrics are shown as heatmaps in Figure 4.2.

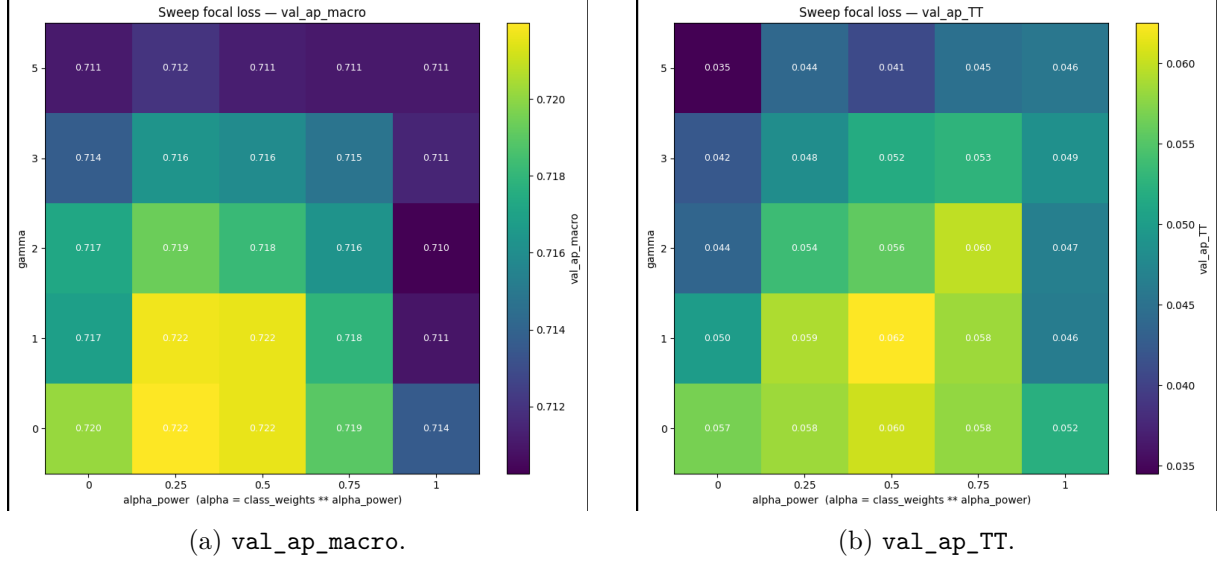


Figure 4.2: Scan of (p_α, γ) for the focal loss. Logbook 24/09; results in `outputs_training/DNN/HH_bttt_dataset/SWEEP_FOCAL_PARAMS_2layers_32_16\_dots` / (one sub-folder per point, e.g. `focal_a0.5_g1`).

- The best region is at low γ (0–1) and $p_\alpha \approx 0.25$ –0.5: the maximum `val_ap_macro` is 0.722 and the maximum `val_ap_TT` is 0.062, at $(p_\alpha, \gamma) = (0.5, 1)$, which is the chosen working point.
- The trend is monotonic in γ : performance degrades as γ grows, more strongly for TT. Since $\gamma = 0$ is a weighted cross-entropy, the modulating factor is here counter-productive. A possible explanation (not tested) is a “double focusing”: TT is already emphasised by the weights and the modulation concentrates the gradient on too few hard examples, making the optimisation noisier.
- The two metrics react very differently. `val_ap_macro` varies only from 0.710 to 0.722 ($\sim 1.7\%$ relative), compatible with the training noise, whereas `val_ap_TT` varies from 0.035 to 0.062 ($\sim 77\%$). The parameters thus matter for the critical class, but barely for the average behaviour. For reference, the TT fraction of the test set is $614/81\,083 \simeq 0.8\%$, so 0.062 is about ten times the random-classifier value, but still small in absolute terms.

4.3.3 Best model

The best configuration is the weighted focal loss with $p_\alpha = 0.5$ and $\gamma = 1$ (run `focal_a0.5_g1`), selected on the validation `ap_macro` (0.7216; best epoch 42 of 50, validation loss 0.172). The network ($21 \rightarrow 32 \rightarrow 16 \rightarrow 4$, ReLU, layer normalisation after each hidden layer) has 1396 trainable parameters. It is trained with AdamW (learning rate 10^{-3} , weight decay 10^{-4}), a `ReduceLROnPlateau` scheduler (factor 0.7, patience 5), at most 50 epochs, early-stopping patience 10 and seed 42. With $p_\alpha = 0.5$ the class weights of `??` are raised to the power 0.5: the weight of TT goes from 38.2 to about 6.2 (and the others to $\approx 1.6, 0.6, 1.2$ for FF, FT, TF). The validation average precision per class is 0.871 (FF), 0.997 (FT), 0.956 (TF) and 0.062 (TT). The training-set metrics refer to $376\,832 = 184 \times 2048$ pairs, i.e. the last incomplete batch of 735 pairs of the 377 567 pairs of the training set was dropped and is not used in the evaluation (`drop\_last` in the training loader).

Table 4.4 gives the test-set metrics and Table 4.5 the one-versus-rest working points obtained by scanning the score of each class (background rejection at fixed efficiency of the class); the plots are in Figure 4.3.

Table 4.4: Test-set metrics of the best model (focal loss, $(p_\alpha, \gamma) = (0.5, 1)$), class assigned with the **argmax** of the output.

Class	Support	Precision	Recall	F_1
FF	8 140	76.15%	87.37%	0.8138
FT	59 204	98.15%	97.42%	0.9778
TF	13 125	91.57%	89.73%	0.9064
TT	614	18.03%	3.58%	0.0598
Macro average		70.98%	69.53%	0.6895
Accuracy	81 083		94.45%	

Table 4.5: Background rejection of the one-versus-rest working points of the best model on the test set, at fixed efficiency of the class.

Class	Class efficiency					
	70%	75%	80%	85%	90%	97%
FF	86.3	68.7	54.1	39.3	26.3	8.5
FT	237.8	191.9	140.3	104.2	66.7	22.2
TF	220.6	179.8	145.8	106.0	62.2	12.2
TT	7.0	6.0	4.9	3.9	2.9	1.3

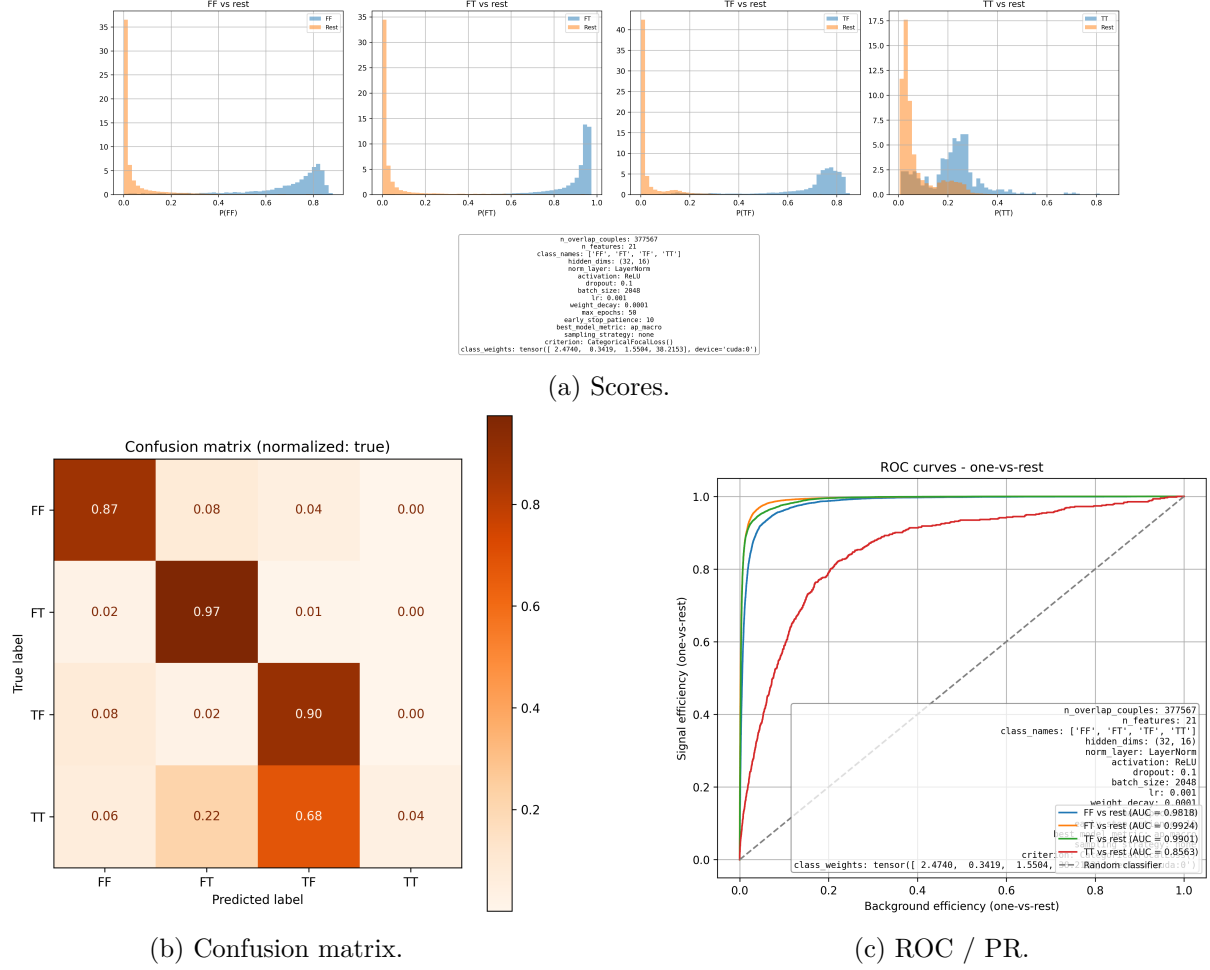


Figure 4.3: Best model, focal loss with $(p_\alpha, \gamma) = (0.5, 1)$. Folder `outputs_training/DNN/HH_bbtt_dataset/SWEEP_FOCAL_PARAMS_2layers_32_16\_dots/focal_a0.5_g1`.

The main features of this working point are the following.

- The global performance is as good as that of the weighted cross-entropy: accuracy 94.5% (against 94.8%) and macro- F_1 0.690, higher than both the cross-entropy (0.677) and the $\gamma = 2$ run (0.646) of Table 4.3. Training and test metrics are compatible (for example the TT recall is 3.8% and 3.6%), so there is no sign of over-fitting.
- For TT, the `argmax` assignment is rare and rather pure: recall 3.6%, precision 18.0%, i.e. about 22 true TT among about 120 predicted ones (derived from the table). This is the opposite trade-off of the $\gamma = 2$ run (recall 42.7%, precision 4.7%). The “best” model is the best on `ap_macro`; it is not the one with the highest TT recall.
- The three classes with enough statistics are well separated by the score of the class: for FT and TF the background rejection is above 100 up to 85% efficiency. For TT it is not: at 70% efficiency 14% of the other pairs pass the cut, which, with a TT fraction of 0.76%, means a purity of only about 4% (derived).
- The TT thresholds are very low (0.18 at 70% efficiency), as the output probability of this class is rarely large.

4.3.4 Transformer

An alternative architecture, `SetTransformerTagger` (module `models.py`, instantiated by `create_model` from the model configuration), was implemented in `notebooks/Transformer_training.ipynb`, after studying the Transformer architecture (self-attention and multi-head attention). The idea is to treat a pair as a *set* of a few objects rather than as a flat vector of 21 numbers, so that the attention can learn explicitly the interaction between them, on a smaller scale than what a tagger such as GN2 does with tracks. The preprocessing is unchanged: the same `.npz` files of the dense network are used.

Architecture. The 21 features of Table 4.1 are partitioned (`feature_groups`) in four groups, each of which becomes a token:

1. the relative variables of the pair (features 1–4: ΔR , $\Delta\eta$, $\Delta\phi$, p_T ratio);
2. the jet (features 5–10, including the GN2 score);
3. the τ (features 11–19, including the three τ scores);
4. the variables derived from the missing transverse momentum (features 20–21: `tau_met_proj`, `tau_mt`).

The data flow is the following.

1. Each token is embedded by a dedicated small network (linear layer, layer normalisation, GELU) that projects the raw features of its group, of different sizes, into the common space of dimension d_{model} ; no padding is needed.
2. A learnable *type embedding* is added to each token, so that the network can tell the four tokens apart.
3. A `TransformerEncoder` with n_{layers} layers (multi-head self-attention and a feed-forward block of dimension $4d_{\text{model}}$, GELU, dropout) lets the tokens exchange information.
4. The four output tokens are averaged (mean pooling, which is invariant to the order of the tokens), and a small head (layer normalisation, linear layer, GELU, dropout, linear layer) returns the four class logits.

The configuration used is $d_{\text{model}} = 32$, 4 heads, 2 layers and dropout 0.1, for 27 844 trainable parameters, about 20 times those of the dense network (1 396). The optimiser, scheduler, batch size, early stopping, model selection on `ap_macro`, seed and class weights are the same as for the dense network (AdamW, learning rate 10^{-3} , weight decay 10^{-4} , batch size 2048, patience 10, seed 42; $\omega_{TT} = 38.2$), with no resampling. The outputs are in `outputs_training/Transformer/HH_bbtt_dataset/` and the repository commit of the run is `c76f0cf`.

Runs. Two trainings were done, with the same loss parameters already used for the dense network in the comparison of ??:

- weighted cross-entropy;
- weighted focal loss with $\gamma = 2$ and full inverse-frequency weights ($p_\alpha = 1$).

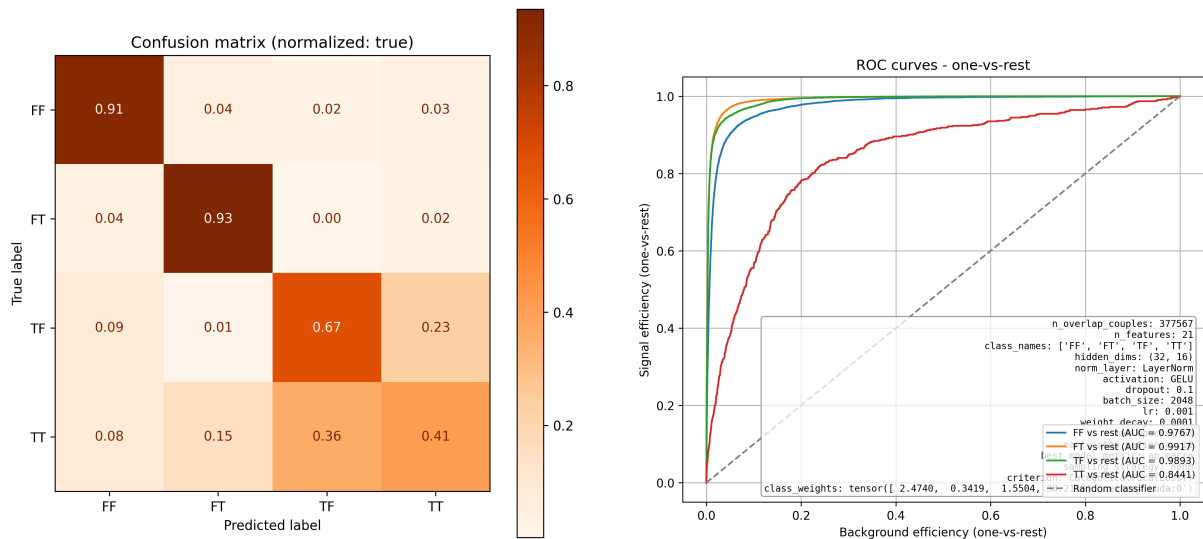
The run reported below is the focal-loss one; training stopped at epoch 32 and the best model, selected on the validation `ap_macro`, is the one of epoch 21 (`ap_macro` = 0.7140, validation loss 0.241).

Table 4.6 compares the test-set metrics with those of the dense network trained with the same loss (Table 4.3), and Figure 4.4 shows the confusion matrix and the ROC curves.

Table 4.6: Test-set metrics of the Transformer and of the dense network, both with the weighted focal loss with $\gamma = 2$, class assigned with the `argmax` of the output.

Class	Precision		Recall		F_1	
	Transf.	DNN	Transf.	DNN	Transf.	DNN
FF	67.91%	66.60%	90.98%	91.31%	0.7777	0.7703
FT	99.00%	99.01%	93.44%	92.80%	0.9614	0.9580
TF	94.55%	95.61%	67.07%	64.43%	0.7848	0.7698
TT	5.05%	4.69%	41.04%	42.67%	0.0899	0.0845
Macro avg	66.63%	—	73.13%	—	0.6535	0.646
Accuracy	88.53%	—	87.7%	—	—	—

[TODO: fill the macro-averaged precision and recall of the dense network with $\gamma = 2$ from its run report]



(a) Confusion matrix (normalised to the true class).

(b) One-versus-rest ROC curves.

Figure 4.4: Transformer (focal loss, $\gamma = 2$) on the test set. Files `confusion_matrix_test_transformer\*.pdf` and `roc_curve_test_transformer\*.pdf` in `outputs_training/Transformer/HH_bbtt_dataset/`; the loss curves (`loss_transformer\*.pdf`) and the score distributions (`score_distribution_test_transformer\*.pdf`) are in the same folder.

Results. With the same loss, the Transformer behaves like the dense network, with a small improvement that is not significant given the statistics of the TT class (614 test pairs).

- Accuracy (88.5% against 87.7%) and macro- F_1 (0.654 against 0.646) are slightly higher, and the TF recall increases by 2.6 points; the TT recall is the same within a few points (41.0% against 42.7%), with the same low purity (5.1%: about 250 true TT among about 5 000 predicted).
- Both models are below the best dense model of Table 4.4 ($(p_\alpha, \gamma) = (0.5, 1)$, accuracy 94.5%, macro- F_1 0.690, validation `ap_macro` 0.7216 against 0.7140): the difference is due to the loss parameters, not to the architecture.
- The confusion matrix shows where the TT predictions come from: 23% of the true TF are classified as TT, which, given the size of the TF class, dominates the TT predictions, and

the true TT are classified as TF in 36% of the cases. A TT pair differs from a TF pair only by the true τ , which is the genuine difficulty of the problem.

- The ROC of TT is the worst, with $\text{AUC} = 0.844$ (against 0.977 for FF, 0.992 for FT, 0.989 for TF). The background rejection at fixed class efficiency is similar to that of the best dense model for FT and TT (TT at 70% efficiency: 6.8 against 7.0) and lower for FF (75.0 against 86.3) and TF (207 against 221).
- The validation loss has its minimum at epoch 15 and then grows while the training loss decreases, and the TT recall is 48.8% on the training set and 41.0% on the test set: a mild over-fitting, with 20 times more parameters than the dense network.

With the present features and statistics the capacity of the model does not seem to be the limiting factor, and the dense network is preferred for its simplicity. The comparison is limited: the Transformer was trained only with one choice of architecture hyper-parameters, and not with the parameters of the best dense model. *Next step should be to re-run the Transformer with the loss parameters of the best dense model ($p_\alpha = 0.5$, $\gamma = 1$) for a like-for-like comparison with Table 4.4 and to try more architecture hyper-parameters of the Transformer (d_{model} , number of heads and layers, dropout, learning rate) and, if useful, a learnable class token instead of the mean pooling.*

4.3.5 Conclusions and open points

- A dense network with few parameters learns FT, TF and FF well with weighted cross-entropy (accuracy 94.8%), but it never predicts TT.
- The focal loss makes TT predictable (recall $\sim 43\%$) at the price of the TF and FF performance, and the TT purity stays very low.
- Within the focal loss, low γ is preferred; with $(p_\alpha, \gamma) = (0.5, 1)$ as best compromise. The effect on the average metric is within the noise, the effect on TT is not.
- *Open:* the TT statistics is the limiting factor: a sizeable part of the TT and FF pairs is irreducible (Section 2.3.1), and more data or a different treatment of the class would be needed.
- *Open:* the TT class merges sub-populations that have not been separated (Chapter 2).
- *Open:* results of the Transformer, and the comparison of the network output with the simple combination of the two offline discriminants (the manual combination of Objective 3.2).

Chapter 5

Objective 3.4 — Signal versus background

5.1 Objective

The original goal of the objective is to study the separation between signal and background events by counting the objects selected by the b -jet and τ discriminants. At the time of writing, the part of the work already done concerns the preliminary step: repeating on the fully hadronic $t\bar{t}$ background the characterisation done on the signal (Objectives 3.1–3.3), to understand whether the overlap, the discriminating variables and the tagger behaviour found on $HH \rightarrow b\bar{b}\tau\tau$ carry over to the background. [The counting of the selected objects is *not* done yet](#) (see the open points).

5.2 Approach

The same chain of scripts of Objectives 3.1–3.3 is applied to the $t\bar{t}$ sample of Table 1.1 (branch `AnalysisMiniTree`), selecting the background sample in `params.py`. Outputs are written to separate folders, so that signal and background can be compared plot by plot:

- overlap characterisation and tagger performance: `outputs_datasets_analysis/outputs_tt/` (same sub-folders as the signal folder `outputs_HH_bbtatau`);
- evaluation of the networks of Chapter 4 on the background: `outputs_training/DNN/EVAL_bkg_tt/`.

Both the pairing with all the pairs below the ΔR threshold and the greedy matching without repetition (`greedy_match_no_reps_indices`, Chapter 2) are run, to check whether the class composition is robust.

Two additional ingredients are used to interpret the results.

- A rough *branching-ratio estimate* of the expected probability that a τ candidate in a (jet, τ) pair is a true hadronic τ , for b -jets and for non- b jets (Section 5.3.2).
- A comparison, for the same variables, of the distributions of the four truth categories on signal and on background.

5.3 Results and conclusions

5.3.1 Composition of the overlapping pairs

In the fully hadronic $t\bar{t}$ sample the minority categories are not TT and FF as for the signal, but TT and FT, i.e. the ones with a true hadronic τ . This is expected: the sample comes from $t\bar{t} \rightarrow WbWb$ with $W \rightarrow q\bar{q}'$ only, so the true τ can only appear in the long decay chains

of heavy-flavour hadrons ($B \rightarrow \tau\nu_\tau X$ from beauty, or $D_s \rightarrow \tau\nu_\tau$ from charm). Table 5.1 gives the composition for $\Delta R < 0.4$ with the two pairing methods of Objective 3.1: all the pairs, with possible repetitions of the objects, and the greedy matching without repetitions (`greedy_match_no_reps_indices`).

Table 5.1: Truth categories of the (jet, τ) pairs with $\Delta R < 0.4$ in the $t\bar{t}$ background, with possible repetitions of the objects and with the greedy matching without repetitions. Counts from the legends of the pair-kinematics plots (logbook 26/09); the fractions of the no-repetition case are computed from the counts.

Class	Meaning	With repetitions		No repetitions	
		N	Fraction	N	Fraction
TF	true b , fake τ	24 896	24.30%	24 763	24.27%
FT	fake b , true τ	72	0.070%	68	0.067%
TT	both true	632	0.617%	632	0.620%
FF	both fake	76 868	75.02%	76 553	75.04%
Total		102 468		102 016	

The two methods give almost the same composition. The greedy matching removes 452 pairs (0.4%), almost all of them FF and TF; the TT count is *unchanged* (632) and the FT count drops by 4 pairs ($72 \rightarrow 68$, within the statistical uncertainty of such a small class). The minority classes are therefore an irreducible fraction of the pairs, as for the signal (Chapter 2), and not an artefact of the repeated use of the same jet or τ in several pairs. The class counts of the two methods are written in the legend of every pair-kinematics plot in `outputs_datasets_analysis/outputs_tt/`.

5.3.2 Comparison with the branching-ratio expectation

From Table 5.1, the probability that the τ of a pair is a true hadronic τ is

$$\text{jet is } b : \quad 632/25\,528 \simeq 2.5\%, \quad (5.1)$$

$$\text{jet is not } b : \quad 72/76\,940 \simeq 0.094\%, \quad (5.2)$$

with a ratio of about 26. These numbers are compared with an order-of-magnitude estimate from branching ratios.

- b -jets: $\text{BR}(b \rightarrow \tau^- \bar{\nu}_\tau X) \approx 2.4\%$ times the hadronic fraction of the τ ($\approx 65\%$) gives $\simeq 1.6\%$.
- Non- b jets: the main source is $W \rightarrow c\bar{s}$ followed by $c \rightarrow D_s$ ($\approx 8\%$) and $D_s \rightarrow \tau\nu$ ($\approx 5.4\%$), times the hadronic fraction of the τ , with a fraction of c -jets among the non- b jets of about 20% (assumed): $\simeq 0.05\%$.

The expectations (1.6% and 0.05%) and the observed values (2.5% and 0.094%) are of the same order of magnitude, with the same hierarchy. This estimate is a plausibility check, not a measurement: [the input rates \(\$c\$ -jet fraction, \$c \rightarrow D_s\$ probability\) are uncertain](#) and the FT class has only 72 pairs.

5.3.3 Discriminating variables on signal and background

Repeating on the background the analyses done for the signal, the variables that discriminate the truth categories on $HH \rightarrow b\bar{b}\tau\tau$ in general *do not* show the same distributions on $t\bar{t}$ (Figure 5.1). This is consistent with the fact that the physical processes ending up in the same class (e.g. FF) are not the same in the two samples. In many other variables (transverse momenta, relative angular variables, ΔR , τ decay mode and prong), the very low statistics of the `_T` categories on

the background makes the comparison with the signal inconclusive; a better knowledge of the physics of these pairs could help to predict the expected behaviour.

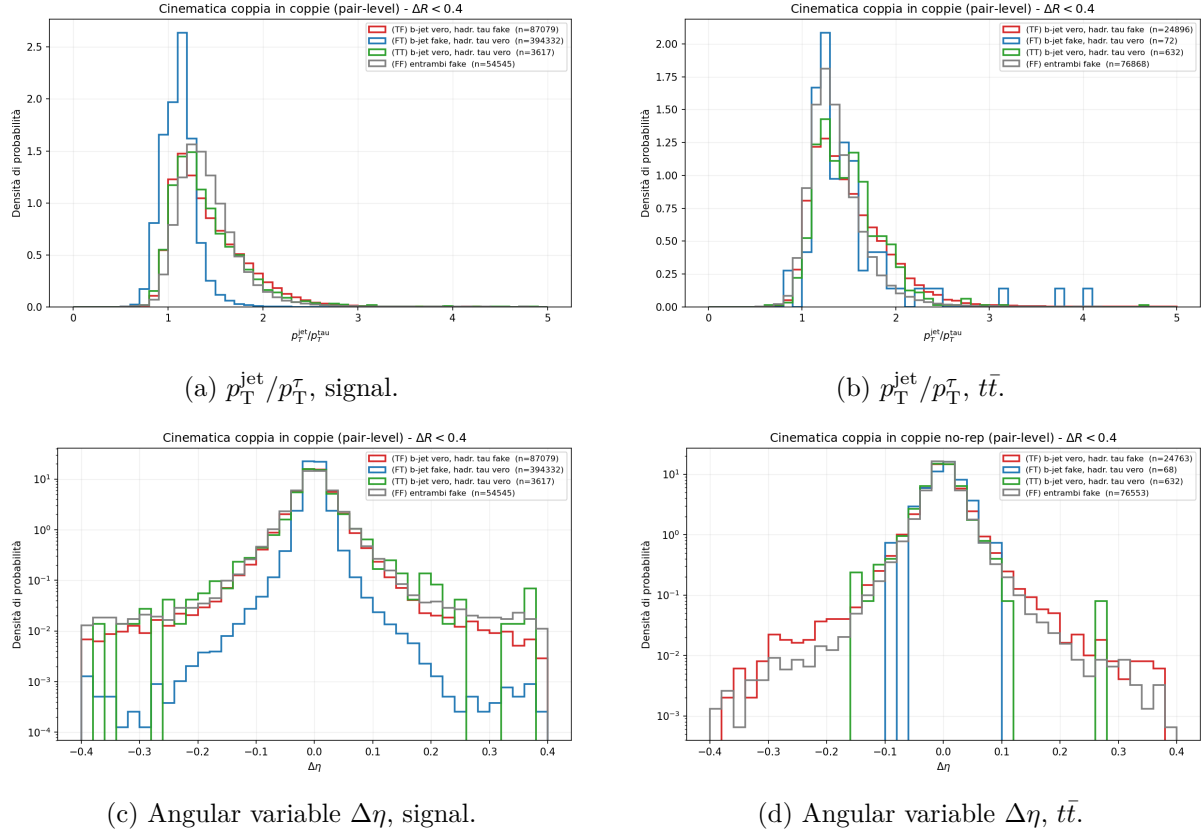


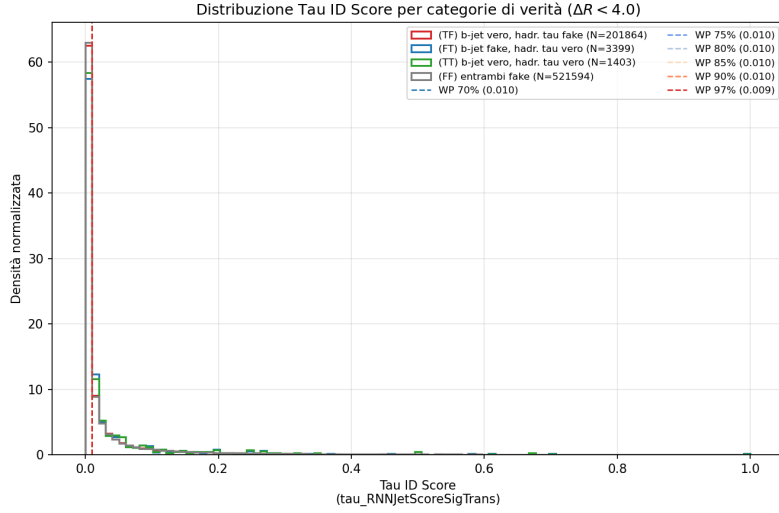
Figure 5.1: Same variables for the four truth categories on signal (left) and $t\bar{t}$ background (right). Logbook 26/09; other variables are in `outputs_datasets_analysis/outputs_tt/discriminance_analysis` and in `outputs_datasets_analysis/outputs_HH_bbtatau/discriminance_analysis`.

5.3.4 Tagger scores on the background

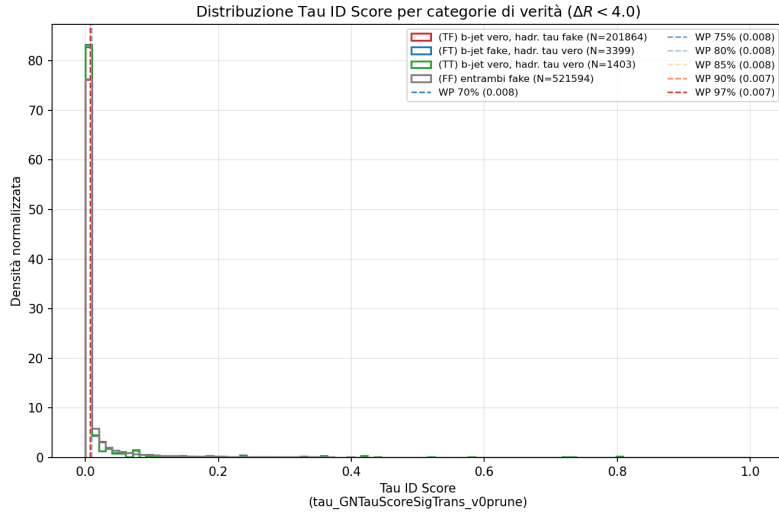
On the signal, the true τ in FT had uniform (signal-transformed) scores, while the rare TT category had low scores (Chapter 3). On the $t\bar{t}$ background the picture is different.

- **τ taggers:** all the events with a true hadronic τ (τ_{had}) are *not* recognised as τ by the RNN (nor, similarly, by GNTau) (Figure 5.2a).
- **GN2:** the difference of the GN2 behaviour between TT and TF is smaller than on the signal (Figure 5.2b).
- **RNNEle:** as on the signal, it does not separate the classes (Figure 5.2c).

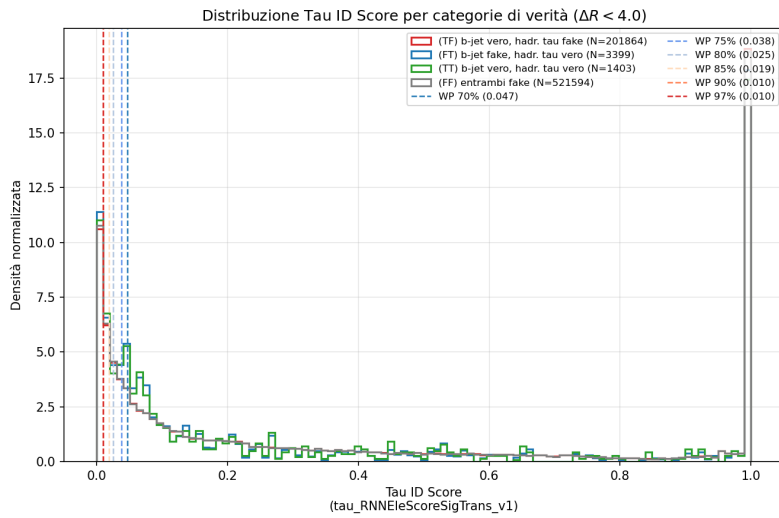
This is compatible with the origin of these τ : they are part of long decay chains of heavy-flavour hadrons and are not isolated, while the τ taggers are trained to separate *prompt* τ from hadronic jets (this statement must be checked in the documentation of the training samples of the taggers).



(a) RNNJet score.



(b) GN2 quantile.



(c) RNNEle score.

Figure 5.2: Tagger outputs for the four truth categories on the $t\bar{t}$ background. Logbook 26/09; same plots as in Figure 3.2 for the signal.

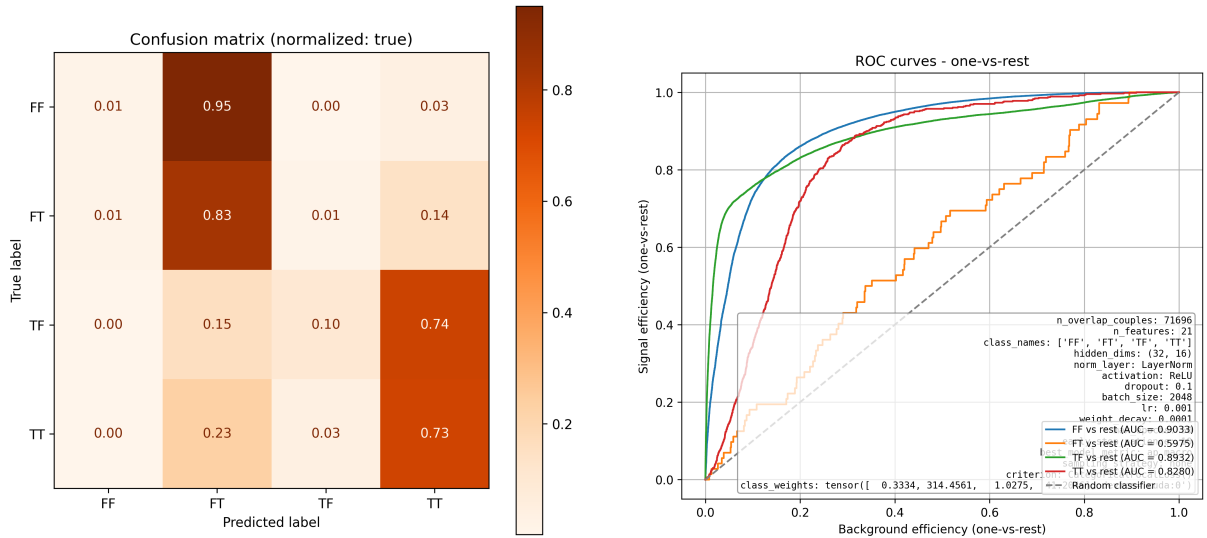
5.3.5 Evaluation of the signal-trained networks on the background

The networks of Chapter 4, trained on the signal sample, were applied to the $t\bar{t}$ background to check whether what they learned carries over. No training or fine-tuning is done on the background: the whole background dataset (102 468 pairs, the same as in Table 5.1, with the same 21 features and standardisation) is used only for inference. The evaluation reported here is for the dense network trained with the weighted focal loss with $\gamma = 2$ and $p_\alpha = 1$, i.e. the same configuration as in the comparison of ???. The equivalent evaluation was done also for the weighted cross-entropy network of that comparison, which gives the same qualitative picture. For completeness, we should evaluate the best model on the signal ($p_\alpha = 0.5$, $\gamma = 1$, Table 4.4) and the Transformer; the result below is already explicit about the lack of generalisation, but the best model is the one that should be quoted. The outputs are in `outputs_training/DNN/EVAL_bkg_tt/`, and the code used is the same evaluation notebook as for the signal, run on `datas/np_arrays/bkg_tt`.

Table 5.2 and Figure 5.3 show that the network does *not* generalise to the background.

Table 5.2: Metrics of the dense network (weighted focal loss, $\gamma = 2$, $p_\alpha = 1$), trained on the signal, on the $t\bar{t}$ background, class assigned with the `argmax` of the output, and area under the one-versus-rest ROC curve of each class.

Class	Support	Precision	Recall	F_1	AUC
FF	76 868	97.04%	1.32%	0.0261	0.903
FT	72	0.08%	83.33%	0.0016	0.598
TF	24 896	93.51%	10.01%	0.1808	0.893
TT	632	2.14%	73.42%	0.0416	0.828
Macro average		48.19%	42.02%	0.0625	
Accuracy	102 468		3.94%		



(a) Confusion matrix (normalised to the true class).

(b) One-versus-rest ROC curves.

Figure 5.3: Evaluation on the $t\bar{t}$ background of the network trained on the signal (focal loss, $\gamma = 2$, $p_\alpha = 1$). Files `confusion_matrix_eval\dots_EVAL_bkg_tt.png` and `roc_curve_eval\dots_EVAL_bkg_tt.png` in `outputs_training/DNN/EVAL_bkg_tt/`; the score distributions (`score_distribution_eval\dots`) are in the same folder.

The main findings are the following.

- The accuracy is 3.9% and the macro- F_1 0.063 (against 87.7% and 0.646 on the signal test set for the same loss, Table 4.3): the classification is wrong almost everywhere. The confusion matrix shows a systematic displacement of the classes: 95% of the true FF (the 75% majority of the background) are assigned to FT, which is the majority class on the signal (73% of the training set), and 74% of the true TF are assigned to TT. As a consequence, only about 1% of the FF and 10% of the TF are recovered, with a very high precision (97% and 94%), simply because very few pairs are assigned to these classes.
- The assignment to the two classes with a true τ is dominated by false positives: the recall of FT and TT is high (83% and 73%) but the precision is 0.08% and 2.1%, with about 7.5×10^4 pairs predicted as FT and about 2.2×10^4 as TT (derived from precision and recall), against 72 and 632 true pairs.
- The ranking power of the output is not completely lost. The AUC is 0.90 for FF, 0.89 for TF and 0.83 for TT, well above the random value, and the background rejection at fixed class efficiency is still non trivial for TT (5.2 at 70% efficiency, 3.5 at 85%) and TF (20.7 and 4.3). It is poor for FT (AUC 0.60, rejection ~ 1.9 at 70%), which is also the class with the smallest statistics (72 pairs). The scores of the true class are, however, peaked at values different from those on the signal, so the thresholds of the signal cannot be used on the background.

The interpretation is a *domain shift*: the network was trained on class priors and on feature distributions of the signal, which are not those of the background. The prior is very different: the FT class is 73% of the signal pairs and 0.07% of the background pairs, and the softmax output carries this prior, which explains why the FF pairs fall in FT. The features that discriminate the classes on the signal do not discriminate on the background in the same way (e.g. the p_T ratio and the angular variables, Figure 5.1), and the τ taggers do not recognise the true τ of the background (Figure 5.2); the network relies on all of them. This effect is not tested here separately.

The conclusion is that a network trained on the signal cannot be used directly on the background, and the evaluation is a negative result relevant for the signal-versus-background goal of Objective 3.4: the truth category of a pair is not the quantity that separates signal from background, and a classifier of the category is not transferable between the two samples. [A dataset that includes background pairs in the training, or a network trained directly to separate signal from background, would be needed.](#)

5.3.6 Conclusions and open points

- On the fully hadronic $t\bar{t}$ background the τ categories are the minority ($\sim 0.7\%$ of the pairs between TT and FT), the opposite of the signal, and a part of them is irreducible, as for the signal.
- Their size is of the same order as expected from branching ratios of B and D_s decays into τ .
- The signal features that discriminate the categories do not automatically do so on the background, and the τ taggers do not recognise the true τ of the background. Both facts limit the direct transfer to the background of the models trained on the signal (Chapter 4).
- *Open (main goal of the objective)*: the counting of the objects selected by the b -jet and τ discriminants, for signal and background, and the resulting signal/background separation, have not been done yet.
- *Open*: check of the τ ancestors on the background and of the training samples of the τ taggers.

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